

Adverse Selection in Cryptocurrency Markets

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Abstract

This paper investigates the influence that information asymmetry may possess upon the future volatility, liquidity, market toxicity and returns within cryptocurrency markets. We use the adverse selection component of the effective spread as a proxy for overall information asymmetry. Using order and trade data from the Bitfinex Exchange, we first document statistically significant adverse selection costs for major cryptocurrencies. Our results also suggest that adverse selection costs, on average, correspond to ten percent of the estimated effective spread, indicating an economically significant impact of adverse selection risk on transaction costs in cryptocurrency markets. We finally document that adverse selection costs are important predictors of intraday volatility, liquidity, market toxicity, and returns.

Keywords: Cryptocurrencies; Market Microstructure; Adverse Selection; Informed Trading.

JEL: G12, G14, G15, C51

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1 Introduction

The notion of fundamental value for cryptocurrencies can be considered to be very different from other financial assets such as stocks or fiat currencies. Cryptocurrencies have been found to provide transactional benefits that money issued by central banks does not (Biais et al., 2020). For example, cryptocurrencies may provide a platform to allow for the completion of transactions even when some centralised currencies or banking systems face significant problems, while cryptocurrencies can also provide flexibility to investors when transferring funds from one country to another, even considering the presence of closed economies. Cryptocurrencies also lead to transaction costs in the form of limited convertibility, a lower rate of acceptance, and fees embedded within the mining process (Biais et al., 2020)¹. The theoretical framework presented in Biais et al. (2020) indicates the existence of a feedback loop from future prices to transactional benefits and current prices; where investors who expect future prices to be high, anticipate elevated transactional benefits, which results in high prices today. Biais et al. (2020) underline that the fundamental value of a cryptocurrency is, therefore, a function of net transactional benefits that depend on its future price. In practice, those with internal knowledge with regards to the technological processes surrounding both cryptocurrency creation and the mining process may better assess net transactional benefits and, therefore, could possess an informational advantage when attempting to understand the fundamental value of the associated product. Therefore, investors who lack sufficient knowledge about the underlying cryptocurrency product might face significant adverse selection problems. Such issues can also arise for both retail and institutional investors of cryptocurrency. Institutional investors, who are traditionally considered to be informed because of their ability to monitor market conditions and process information at lower costs (Seppe, 1992; Hessel and Norman, 1992; Lang and McNichols, 1997), have become increasingly prevalent in cryptocurrency markets in recent years (Borri, 2019)². While earlier attempts by Momtaz (2020) and Chod and Lyandres (2020) focus on information asymmetry at the initial coin offering stage, the role of information asymmetry in secondary cryptocurrency markets remains unexplored.

This research aims to complement the existing debate by examining the impact of information

¹See also Easley et al. (2019) for a game-theoretic view on inter-dynamics of the cryptocurrency prices, transaction fees, mining rewards and waiting times.

²Cryptocurrencies are even included in portfolios of several hedge funds (See the [link](#)).

asymmetry at the exchange level when considering the volatility, liquidity, market toxicity, and returns of cryptocurrency. We proxy information asymmetry with the adverse selection component of the effective spread. Specifically, we make use of the seminal models of [Glosten and Harris \(1988\)](#), [Huang and Stoll \(1997\)](#), and [Madhavan et al. \(1997\)](#) to decompose the spread into two parts: an adverse selection component related to permanent price change due to informed trading, and a transitory component related to the temporary price change and order-processing costs. We also apply the three-way decomposition of [Huang and Stoll \(1997\)](#) to account for the changes in the transaction costs due to inventory risk. In a panel regression framework, we attempt to specifically investigate the following questions: (i) does an increase in the adverse selection component of the spread significantly impact the future volatility of cryptocurrency returns? (ii) does the adverse selection component of the spread influence cryptocurrency liquidity? (iii) is the adverse selection component related to a reduction in market toxicity? And finally, (iv) can the adverse selection component act as a significant predictor of short-term cryptocurrency returns?

Accordingly, using the limit order book and trade data sourced from the Bitfinex Exchange, we document that the adverse selection component of the effective spread is significant for the twelve most frequently traded cryptocurrencies between August 2017 to June 2018. Adverse selection costs, on average, correspond to ten percent of the estimated spread, suggesting that the proportion of the spread attributable to adverse selection risk in cryptocurrency markets is also economically significant. Moreover, we show that the adverse selection component of the spread is a significant predictor of future volatility proxied by the standard deviation of cryptocurrency returns. Our results suggest that informed trading activity increases the future levels of return volatility in cryptocurrency markets. Furthermore, we present evidence that indicates the existence of an inverse relationship between adverse selection costs and cryptocurrency liquidity. Through intraday panel regressions, we initially show that informed trading activity increases the realized spread³ for cryptocurrency trading. We then document that an increase in adverse selection costs results in higher levels of order-book slope [Amaya et al. \(2018\)](#) and [Amihud \(2002\)](#) illiquidity levels. However, our findings indicate that in cryptocurrency markets, the adverse selection component of the spread, does not have a statistically significant impact on future levels of order-book depth. The impact of adverse

³We take realized spread as the average value of the difference between the best ask and bid price at each time tick in the selected time interval for the selected cryptocurrency.

selection costs on future liquidity is the same for both the bid- and ask-side of the order book, overruling a potential argument that has been based on asymmetric impact. The impact of adverse selection costs on future levels of volatility and liquidity is similar to the previous findings documented when investigating currency market behaviour (Payne, 2003; Frömmel et al., 2008); while partially contradicting prior evidence based on stock markets interactions (Rzayev and Ibikunle, 2019). Therefore, these findings can also contribute, albeit with a limited capacity, to the current debate on discussions with regards to which asset class (whether stock, commodity, or currency) to which cryptocurrencies belong. Results indicate that such products appear to be best described as a currency when considering other alternatives.

Further, we show that the relationship between adverse selection costs and future market toxicity levels is negative and statistically significant. This result might imply that price discovery due to informed trading activity can actually lower the future profits for trading based on private information, which in turn would result in lower toxicity levels. In line with this result, we finally reject the hypothesis that adverse selection cost is not a significant predictor of cryptocurrency returns when considering data at a one-hour time interval. In particular, we document that an increase in adverse selection costs significantly reduces future returns. The direction of the relationship between informed trading and future returns is consistent with the previous findings presented by Rzayev and Ibikunle (2019). Our findings suggest that cryptocurrency markets are relatively efficient within a short time horizon. When considering transaction costs, the direction of the relationship between informed trading and cryptocurrency returns is inconsistent with the expectations of Biais et al. (2020). However, the impact of adverse selection can be considered negligible compared to significant events that indicate the easiness of transactions with cryptocurrencies⁴.

Our paper contributes to the growing literature on the microstructure of cryptocurrency markets⁵. In particular, to the best of our knowledge, this study is the first to explore the intraday relationships between informed trading, transaction costs, liquidity, volatility, and returns by using a cross-section of cryptocurrencies traded in the same market. Therefore, our results can shed light on the main determinants of trading activity and price discovery in cryptocurrency markets which are of direct

⁴Such as developments regarding the convertibility of cryptocurrencies with other currencies, introduction or shutdown of large platforms, hacking activities, etc. (Biais et al., 2020)

⁵See Akyildirim et al. (2020); Alexander and Heck (2020); Alexander et al. (2020) and Alexander et al. (2020) as recent empirical studies on the microstructure of some cryptocurrencies and cryptocurrency derivatives.

importance to cryptocurrency traders and market regulators from various perspectives.

The rest of this paper is presented as follows: Section 2 provides a concise presentation of our sample data. Section 3 discusses methodology, while Section 4 reports the associated results. Finally, Section 5 concludes.

2 Data and Variables

2.1 Data

For this study, we collect the intraday cryptocurrency data from the Kaiko digital asset store⁶. Specifically, we employ the U.S. dollar-denominated cryptocurrency order book and trade data (which includes the true side of the trade initiator) from the Bitfinex exchange. Even though a large body of work focuses on cryptocurrency markets (Alexander and Dakos, 2020; Bariviera and Merediz-Solà, 2021), only a few studies focus on the intraday dynamics of cryptocurrency trading (Alexander and Heck, 2020; Alexander et al., 2020; Griffin and Shams, 2020; Makarov and Schoar, 2020). Our raw dataset covers the period between 1 April 2013 to 24 June 2018. The starting period of this dataset covers only the intraday trading data for Bitcoin. Throughout our sample period, 77 cryptocurrencies traded against the USD, but only a few have available trading data for almost a year. For those cryptocurrencies, the earliest starting date is 12 August 2017. We, therefore, select the period between 12 August 2017 and 24 June 2018 to best represent one of the most liquid periods for the Bitfinex exchange. Throughout the period between 2019 through 2021, many new cryptocurrency exchanges began to trade, diminishing the market share of Bitfinex, as presented in Figure 1. In the period after July 2018, several events generated substantial issues and reputational damage for Bitfinex, coinciding with several sharp reductions in trading volumes⁷. While Bitfinex later recovered much of its market share, our period of analysis surrounds that in which Bitfinex

⁶For a recent study that also uses this database, see Makarov and Schoar (2020). See Alexander and Dakos (2020) for an extensive critique on different cryptocurrency data sources.

⁷Some events occurred during the period after the conclusions of our selected data, but one of the most serious surrounding the case where iFinex, Inc., the parent company of Bitfinex became the source of a lawsuit instigated by the New York attorney general as a result of a loan it received from Tether to cover more than \$850 million that had disappeared from accounts linked to a previous partner. Global Trade Solutions, AG, which operated as Crypto Capital, and other companies to which iFinex had deposited its funds. Crypto Capital was previously a payment processor for the exchange between 2014 to 2018.

was firmly a world-leading exchange⁸.

INSERT FIGURE 1 HERE

We employ two filtering mechanisms to our raw data to achieve a representative sample. First, we choose cryptocurrencies that have data starting on 12 August 2017 and are still actively traded on the last day of our sample. Second, for any selected frequency⁹, we choose those cryptocurrencies having less than a 1% non-trading time interval within the sample period. With this selection criteria, we are left with the following: Bitcoin Cash (BCH), Bitcoin (BTC), Dash (DSH), EOS (EOS), Ethereum Classic (ETC), Ethereum (ETH), Iota (IOT), Litecoin (LTC), OmiseGO (OMG), Monero (XMR), Ripple (XRP), and Zcash (ZEC).

2.2 Market variables

Using the intraday trades, we compute cryptocurrency-specific measures for volatility, liquidity, and toxicity. Specifically, we calculate the standard deviation of returns (σ^R), realized spread ($SPREAD$), Amihud illiquidity ($ILLIQ$), order-book slope ($SLOPE$), order-book depth ($DEPTH$), market toxicity (MT), total trading volume (TV), and the difference between buyer and seller-initiated trades (BSI) for all sample cryptocurrencies and all of the sampling frequencies. Total volume $TV_{i,t}$ is calculated as the logarithm of the U.S. dollar-denominated total volume of trades for cryptocurrency i in time interval t . Further, return $R_{i,t}$ is defined as $R_{i,t} = \log(P_{i,t}/P_{i,t-1})$ where $P_{i,t}$ is the last price in time interval t for cryptocurrency i , while the volatility of returns, $\sigma_{i,t}^R$ is the standard deviation of returns of the transactions in interval t for cryptocurrency i .

We identify the buy-sell imbalance ($BSI_{i,t}$) as the absolute difference between buyer- and seller-

⁸While focusing on a single exchange for a relatively short time interval might generate concerns with regards to the representativeness of the results, it is essential to underline that in cryptocurrency research, there are significant issues regarding data aggregation from multiple sources, especially for multivariate analyses. [Alexander and Dakos \(2020\)](#) underline problems with timestamp mismatch, non-synchronicity, exchange arbitrage, and the use of non-traded prices when data from multiple sources are used in multivariate cryptocurrency research. To avoid problems associated with non-concurrent prices, we solely focus on the intraday dynamics in a single market, Bitfinex. Even though Bitfinex is a reliable exchange with high liquidity and low counterparty risk ([Borri and Shakhnov, 2022](#)), the selected time interval prevents potential problems previously documented for the data feed from Bitfinex. Specifically, price deviations between Bitfinex and other centralised cryptocurrency exchanges were minimal during our examined sample period. This is mainly because our sample ends before November 2018, when Bitfinex introduced trading on Tether, creating significant discrepancies with prices in different exchanges ([Alexander and Dakos, 2020](#)).

⁹We use three sampling frequencies based on 1-min, 15-min, and 60-min data intervals.

initiated trades in time interval t for cryptocurrency i . That is,

$$BSI_{i,t} = |B_{i,t} - S_{i,t}| \quad (1)$$

where $B_{i,t}$ ($S_{i,t}$) is the number of buyer-initiated (seller-initiated) trades in the time interval t for cryptocurrency i , while market toxicity ($MT_{i,t}$), is defined as follows:

$$MT_{i,t} = \frac{BSI_{i,t}}{(B_{i,t} + S_{i,t})} \quad (2)$$

where $B_{i,t}$ ($S_{i,t}$) is the number of buyer-initiated (seller-initiated) trades in the time interval t for cryptocurrency i . Specifically, we measure liquidity in various ways: [Amihud \(2002\)](#) measure for cryptocurrency i in time interval t is the ratio of absolute stock return to the total trading volume for i ; that is:

$$ILLIQ_{i,t} = \frac{|R_{i,t}|}{TV_{i,t}} \quad (3)$$

Market depth ($DEPTH_{i,t}$) is the average of the average volume waiting at the best bid and ask levels at each time tick in the time interval t for cryptocurrency i , where ask-side depth ($DEPTH_{i,t}^A$) is the average volume waiting at the best ask level at each time tick in the time interval t for cryptocurrency i . Similarly, bid-side depth ($DEPTH_{i,t}^B$) is the average volume waiting at the best bid level at each time tick in the time interval t for cryptocurrency i . We calculated the $SPREAD_{i,t}$ as the average value of the difference between the best ask and bid price at each time tick in the time interval t for cryptocurrency i . The order-book slope ($SLOPE_{i,t}$) is measured as the marginal liquidity cost as in [Amaya et al. \(2018\)](#). In particular, assume that at a give time t , the best N offer prices are given by $p_1 < \dots < p_N$ and the best N bid prices by $p_{-1} > \dots > p_{-N}$. Denote the quantity offered at price p_i by x_i , the total quantity available up to the i th best ask price by X_i , and the average price per share by $\hat{S}_t(X_i)$. Namely,

$$X_i = \sum_{j=1}^i x_j \quad (4)$$

$$\hat{S}_t(X_i) = \frac{\sum_{j=1}^i p_j x_j}{X_i} \quad (5)$$

In a similar way for the bid side, we calculate

$$X_{-i} = \sum_{j=1}^i x_{-j} \quad (6)$$

$$\hat{S}_t(X_{-i}) = \frac{\sum_{j=1}^i p_{-j} x_{-j}}{X_{-i}} \quad (7)$$

Then we apply the following linear regression model

$$\hat{S}_t(X_i) = \hat{\alpha}_t + \hat{\lambda}_t X_n + \epsilon_t, \quad n = -N, \dots, -1, 1, \dots, N. \quad (8)$$

We take the average of $\hat{\lambda}_t$ for each time tick in the time interval t and use it as the order-book slope for cryptocurrency i . Finally, we calculated the bid-side slope ($SLOPE_{i,t}^B$) by applying the above procedure only to the bid-side of the order book. Similarly, the ask-side slope ($SLOPE_{i,t}^A$) is found by applying the above procedure only to the ask-side of the order book.

2.3 Adverse selection measures

In addition to market variables, we compute the three measures for the adverse selection component of the spread $\{ASC_{GH}, ASC_{HS2}, ASC_{MRR}\}$ for all cryptocurrencies in our sample, and for all of the selected sampling frequencies using the seminal works of [Glosten and Harris \(1988\)](#); [Huang and Stoll \(1997\)](#) and [Madhavan et al. \(1997\)](#), respectively. All of these papers have a two-way decomposition of the spread where there is a transitory component related to the order processing costs (OPCs) and a component related to the adverse selection costs (ASCs). Furthermore, [Huang and Stoll \(1997\)](#) document that both [Glosten and Harris \(1988\)](#) and [Madhavan et al. \(1997\)](#) models ignore the role of inventory costs (IC), therefore authors provide a three-way decomposition. ASC_{HS3} denotes the adverse selection costs computed with the three-way decomposition of the [Huang and Stoll \(1997\)](#) framework. Further details regarding these measures are provided in [Section 3](#).

2.4 Descriptive statistics

Tables [1](#) and [2](#) present the descriptive statistics and correlations respectively, among currency-specific measures calculated over 60-minute frequency sampling. The largest positive correlation

is identified amongst the adverse selection measures ASC_{GH} , ASC_{HS2} and ASC_{MRR} with a presented estimated value of almost 100%, whereas the largest negative correlation is identified in the relationship between ASC_{GH} and trading volume, with a correlation coefficient of approximately -50%. The contemporaneous pairwise correlation between TV and the adverse selection component of the spread is identified to be negative and significant for all models. Similarly, we observe a negative and statistically significant correlation between ASC_{GH} , ASC_{HS2} and ASC_{MRR} and σ^R . Moreover, there is a positive correlation between adverse selection costs and illiquidity measured by $SPREAD$, $ILLIQ$, and $SLOPE$. We also observe a negative correlation between adverse selection costs and order-book depth ($DEPTH$). To that end, we suspect that there is an inverse relationship between informed trading activity and liquidity in cryptocurrency markets.

INSERT TABLES 1 and 2 HERE

3 Methodology

3.1 Estimating the adverse selection component of the spread

To compute the adverse selection component of the spread, we rely on the seminal works of [Glosten and Harris \(1988\)](#), [Huang and Stoll \(1997\)](#), and [Madhavan et al. \(1997\)](#). These studies separate price changes into a transitory component related to the order processing costs (OPC), inventory holding costs (IC) (in the case of [Huang and Stoll \(1997\)](#)), and an adverse selection component (ASC) associated with permanent price changes and informed trading. In line with these studies, we define m_t as the true value of the underlying security conditional on the information available at time t . With the information arriving in the market, true value changes with the new trade in the following manner:

$$m_t = m_{t-1} + Z_t Q_t + \epsilon_t \quad (9)$$

where Q_t corresponds to trade direction, that is, $Q_t = 1$ ($Q_t = -1$) if the trade is buyer- (seller-) initiated. Z_t is the adverse selection component that has an impact on the true prices which are unobservable. Instead of the true value of the asset, traders observe the price which is modelled as

follows:

$$P_t = m_t + Q_t C_t \quad (10)$$

Observed price also changes with the new trade but this change is temporary. Therefore C_t can be labelled as the transitory component of the trade which affects observed prices but not the true value. Transitory component of the trade is rather associated with the order processing costs (OPCs). [Glosten and Harris \(1988\)](#) model the adverse selection component and transitory component as a function of trade size V_t , as follows,

$$Z_t = z_0 + z_1 V_t \quad (11)$$

$$C_t = c_0 + c_1 V_t \quad (12)$$

The system of equations (9)-(12) cannot be estimated directly as m_t is not observable. However, using the price difference $\Delta P_t = P_t - P_{t-1}$ we can write the system as:

$$\begin{aligned} \Delta P_t &= m_t + Q_t C_t - m_{t-1} - Q_{t-1} C_{t-1} \\ &= m_{t-1} + Q_t Z_t + \epsilon_t + Q_t C_t - m_{t-1} - Q_{t-1} C_{t-1} \\ &= Q_t Z_t + Q_t C_t - Q_{t-1} C_{t-1} + \epsilon_t \\ &= Q_t(z_0 + z_1 V_t) + Q_t(c_0 + c_1 V_t) - Q_{t-1}(c_0 + c_1 V_{t-1}) + \epsilon_t \\ &= z_0 Q_t + z_1 Q_t V_t + c_0(Q_t - Q_{t-1}) + c_1(Q_t V_t - Q_{t-1} V_{t-1}) + \epsilon_t \\ &= z_0 Q_t + z_1 Q_t V_t + c_0(\Delta Q_t) + c_1(\Delta Q_t V_t) + \epsilon_t \end{aligned} \quad (13)$$

where the implied adverse selection component of the spread is given by $ASC_{GH} = 2 \times (\hat{z}_0 + \hat{z}_1 V_t)$. The order processing costs are associated with the transitory component which is given by $OPC_{GH} = 2 \times (\hat{c}_0 + \hat{c}_1 V_t)$. [Madhavan et al. \(1997\)](#) assume the unexpected portion of the order flow affects the true value of the underlying asset, and authors construct a trade indicator spread model similar to [Glosten and Harris \(1988\)](#). Their model can be written as:

$$\Delta P_t = (\theta + \phi) Q_t - \phi Q_{t-1} + \epsilon_t \quad (14)$$

In this framework, the adverse selection component of the spread is then calculated as $ASC_{MRR} = 2\hat{\theta}$ and the associated order processing costs are given by $OPC_{MRR} = 2\hat{\phi}$. Huang and Stoll (1997) model the change in price as the proportion of the constant half spread ($\frac{\nu}{2}$). They indicate that the observed price (p_t) deviates from the true value of the asset (m_t) by the signed half spread.

$$P_t = m_t + \frac{\nu}{2}Q_t + \epsilon_t. \quad (15)$$

Taking the first difference in the price equation in a similar fashion results in the following model:

$$\Delta P_t = \lambda \frac{\nu}{2}Q_{t-1} + \frac{\nu}{2}(\Delta Q_t) + \epsilon_t \quad (16)$$

where $ASC_{HS2} = \hat{\lambda} = \hat{\alpha} + \hat{\beta}$ and $\hat{\alpha}$ is the percentage of the half spread attributable to adverse selection and $\hat{\beta}$ is the percentage of the half spread attributable to the inventory costs (ICs) in Huang and Stoll (1997) framework. The two-way decomposition of Huang and Stoll (1997) presented in equation (16) is identical with the framework of Madhavan et al. (1997) presented in equation (14). Huang and Stoll (1997) further demonstrate that both Glosten and Harris (1988) and Madhavan et al. (1997) ignore the role of ICs. That is, both $\hat{Z}_t$ and $\hat{\theta}$ assume $\hat{\beta} = 0$. Therefore, authors extend their basic model with induced serial correlation in trade flows to provide a three-way decomposition of the spread consisting of OPC, IC and ASC. Specifically, they show that the conditional expectation of the trade indicator at time $t - 1$, given $t - 2$ as:

$$E(Q_{t-1}|Q_{t-2}) = (1 - 2\pi)Q_{t-2} \quad (17)$$

and modify their original approach as follows:

$$\Delta P_t = \frac{\nu}{2}Q_t + (\alpha + \beta - 1)\frac{\nu}{2}Q_{t-1} - \alpha\frac{\nu}{2}(1 - 2\pi)Q_{t-2} + \epsilon_t \quad (18)$$

Therefore simultaneously estimating the equations (17) and (18) for the three-way decomposition of Huang and Stoll (1997) will yield a separate estimate for the proportion of adverse selection costs ($ASC_{HS3} = \hat{\alpha}$), proportion of inventory costs $IC_{HS3} = \hat{\beta}$ and the proportion of order processing

costs ($OPC_{HS3} = 1 - \hat{\alpha} - \hat{\beta}$) of the estimated spread.

3.2 Panel Regressions

3.2.1 The relationship between informed trading and volatility of cryptocurrency returns

The seminal model of Kyle (1985) provides a theoretical framework for driving equilibrium prices of financial securities under the presence of information asymmetry between traders. The model indicates that information is incorporated into prices at a constant rate. Previous studies show a significant and positive relationship between trading volume and return volatility. For example, Collin-Dufresne and Fos (2016) extend the model of Kyle (1985) and indicate that volatility and return relationship is driven by uninformed trading. In addition, several studies conjecture a negative relationship between informed trading activity and return volatility (e.g., Rzaev and Ibikunle (2019) for U.S. equities). On the contrary, Frömmel et al. (2008) documents a positive relationship between informed trading activity and return volatility for foreign exchange (FX) markets. Specifically, the authors argue that order flows of big financial institutions and banks contribute to explaining FX return volatility. To this end, we examine the predictive relationship between adverse selection costs and return volatility, and we form our first hypothesis using the following model:

$$\begin{aligned} \sigma_{i,t}^R = & \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} \\ & + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \varepsilon_{i,t} \end{aligned} \quad (19)$$

where ASC^k , OPC^k and IC^k respectively denote the adverse selection costs, order processing costs and inventory holding costs calculated using two-way decompositions of Glosten and Harris (1988); Madhavan et al. (1997) and Huang and Stoll (1997) framework and the three-way decomposition of Huang and Stoll (1997), that is, $k \in \{GH, MRR, HS2, HS3\}$. By construction, $IC^k = 0$ for $k \in \{GH, MRR, HS2\}$. α_i and θ_t respectively denote cryptocurrency- and time-fixed effects.

Hypothesis 1: Adverse selection component of the spread is a significant predictor of future return

volatilities in cryptocurrency markets,

$$H_0 : \gamma_6 = 0$$

$$H_A : \gamma_6 \neq 0$$

3.2.2 The relationship between informed trading and cryptocurrency liquidity

In the microstructure literature, there are competing claims regarding the relationship between the adverse selection cost of spread and the liquidity of assets. Huang and Stoll (1997) argued that intermediaries in financial markets face three costs that determine the realised bid-ask spreads: order processing costs, inventory holding costs and adverse selection costs. Glosten and Milgrom (1985) suggest that intermediaries widen the spreads under the risk of being adversely selected when trading against informed traders. To this end, *ceteris paribus*, they expect the realised spread to be positively related to adverse selection costs which in turn reduces the liquidity of assets as the transaction costs increase for an average investor.

On the contrary, in financial markets, informed investors are expected to time their trades during periods when there are large liquidity traders present in the market. Admati and Pfleiderer (1988) indicate that informed investors disguise their information when there are high levels of liquidity trading. Ibikunle (2018) documents that informed trading activity increases with overall trading volume determined by liquidity trading. From the cost perspective, in dealership markets, liquidity trading can provide lower inventory holding costs for intermediaries therefore the impact of informed trading on transaction costs can be minimal.

Cryptocurrency markets are order-driven markets without designated market makers, but liquidity is then typically supplied by voluntary market makers who largely follow the same quotation principles as designated market makers. For most asset classes traded as limit order book markets, the market making is now dominated by high-frequency traders (Nawn and Banerjee, 2019; Chen and Garriott, 2020). It has been widely reported that a significant number of high-frequency trading firms are trading cryptocurrencies¹⁰. To that end, inventory costs can also be significant determinants of overall liquidity. Furthermore, the trading fees at Bitfinex range from 10 to 20 basis points for low-volume traders. This is very high compared to equity exchanges in the U.S. where the trading

¹⁰See <https://www.ft.com/content/40a86de6-b5dd-11e7-a398-73d59db9e399>

fees are typically lower than one basis point. Our panel regressions are presented below to enable us to examine the predictive relationship between adverse selection costs and overall liquidity in cryptocurrency markets, controlling for the order processing and inventory holding costs. This leads us to our second hypothesis:

Hypothesis 2: An increase in the adverse selection component of the spread decreases liquidity in cryptocurrency markets.

$$ILLIQ_{i,t} = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} \quad (20) \\ + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \varepsilon_{i,t}$$

where $ILLIQ_{i,t}$ is the [Amihud \(2002\)](#) illiquidity measure. Similarly, we provide direct test on the effect of informed trading on the transaction costs in cryptocurrency markets by running the following model:

$$SPREAD_{i,t} = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} \quad (21) \\ + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \varepsilon_{i,t}$$

where $SPREAD_{i,t}$ is the realised spread for cryptocurrency i in time interval t . Moreover, we also examine the relationship between adverse selection costs and order-book slope where the order-book slope is defined as the marginal liquidity cost as in [Amaya et al. \(2018\)](#).

$$SLOPE_{i,t} = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} \quad (22) \\ + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \varepsilon_{i,t}$$

Since $ILLIQ$, $SPREAD$ and $SLOPE$ are inverse liquidity measures, Hypothesis 2 can be represented in the following form:

$$H_0 : \gamma_6 = 0$$

$$H_A : \gamma_6 > 0$$

Finally, we examine the relationship between adverse selection costs with depth under the following setting:

$$DEPTH_{i,t} = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} \quad (23) \\ + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \varepsilon_{i,t}$$

where $DEPTH_{i,t}$ is the tick average of the average volume waiting at the best bid and ask levels in the order book in time interval t for cryptocurrency i . For depth, we change the alternative hypothesis as

$$H_A : \gamma_6 < 0$$

3.2.3 The differential impact of adverse selection on bid- and ask-side liquidity

Recently, theoretical studies indicated that liquidity in down markets can be radically different than liquidity in up markets, underlining the importance of differences in bid-side and ask-side liquidity (Roşu, 2009). Empirically, Brennan et al. (2012) document that ask-side illiquidity estimates are priced more strongly than on the bid-side in the cross-section of expected returns. Furthermore, they conclude that sell-side liquidity is the main contributor to the liquidity premium in U.S. markets. Similarly, using eleven years of limit order book data for NYSE, Cenesizoglu and Grass (2018) show that even though the commonality in liquidity on the bid-side is larger than the ask-side during the financial crisis, only the ask side illiquidity predicts future returns.

In this study, using the limit order book of cryptocurrencies, we also compute the bid-side and ask-side liquidity measures separately. The order-book slope for the ask-(bid-)side is given by $SLOPE_{i,t}^A$ ($SLOPE_{i,t}^B$). Similarly, we represent the ask-(bid-)side depth with $DEPTH_{i,t}^A$ ($DEPTH_{i,t}^B$). We respectively use the ask- and bid-side slope and depth measures as dependent variables in equations (22) and (23) to test our second hypothesis. This allows us to examine the potential asymmetric impacts of adverse selection costs on ask- and bid-side liquidity.

3.2.4 The relationship between informed trading and cryptocurrency market toxicity

Order-flow (Market) toxicity is defined as the adverse selection costs faced by investors who are not aware that they provide liquidity at a loss (Easley et al., 2011). Rzayev and Ibikunle (2019)

argue that market toxicity can be considered as a high-frequency equivalent of adverse selection risk. To this end, we expect market toxicity to raise with an increase in the adverse selection costs. Therefore, our third hypothesis can be stated as follows:

Hypothesis 3: An increase in the adverse selection component of the spread increases the market toxicity in the cryptocurrency markets.

$$MT_{i,t} = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} \quad (24)$$

$$+ \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \varepsilon_{i,t}$$

$$H_0 : \gamma_6 = 0$$

$$H_A : \gamma_6 > 0$$

3.2.5 The relationship between informed trading and cryptocurrency returns

The efficient market hypothesis indicates that prices reflect all available information about the value of the underlying financial securities. Prices change only with the arrival of new information and since the arrival of new information is random, it is impossible to predict future prices. There is a large body of literature documenting that prices cannot be predicted consistently over a daily horizon (Fama, 1970). On the other hand, Chordia et al. (2002, 2008) find that lag order imbalances can be a significant predictor of returns over short horizons and argue that investors can be slow to act on new information. Accordingly, order imbalances can proxy for informed trading which affects the true value of the underlying security. Rzaev and Ibikunle (2019) state that the transitory component of the order volume (which is a proxy for informed trading) is a significant and inverse predictor of short-term stock returns which is consistent with the previous literature that documents market inefficiencies over short horizons. To this end, we expect the adverse selection component of the spread to be a significant predictor of short-term cryptocurrency returns.

In addition to the argument above, the theoretical cryptocurrency pricing framework by Biais et al. (2020) indicates that investors, who expect future prices to be high, anticipate future transactional benefits which in turn results in high prices today. Cryptocurrencies also entail transaction costs in the form of limited convertibility, lower rate of acceptances and other problems embedded in the

transaction and mining processes. Accordingly, the authors show that there is a positive (negative) relationship between cryptocurrency returns and transaction costs (benefits). Considering all these arguments above, our fourth hypothesis follows:

Hypothesis 4: Adverse selection component is a significant predictor of short-term cryptocurrency returns.

$$R_{i,t} = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} \quad (25)$$

$$+ \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \gamma_9 DEPTH_{i,t-1} + \gamma_{10} SLOPE_{i,t-1} + \varepsilon_{i,t}$$

$$H_0 : \gamma_6 = 0$$

$$H_A : \gamma_6 \neq 0$$

4 Results

4.1 Estimating the adverse selection component of the spread

We compute the adverse selection component of the cryptocurrency spreads through the use of the models presented by [Glosten and Harris \(1988\)](#); [Huang and Stoll \(1997\)](#) and [Madhavan et al. \(1997\)](#). To test as to whether we observe significant informed trading activity in cryptocurrency markets, we develop upon equations (13), (14), (16), and (18) using intraday trades of all cryptocurrencies in our sample period with ordinary least squares¹¹. Table 3 presents the adverse selection component of the spread estimated by the [Glosten and Harris \(1988\)](#) model. The second column in Table 3 presents the mean estimate for the transitory component (OPC^{GH}) throughout the sample period. The third column corresponds to the p-value for the test with the null hypothesis of $OPC^{GH} = 0$. The fourth and fifth columns respectively correspond to the estimate for the adverse selection costs and the corresponding p-value for the test with the null hypothesis of $ASC^{GH} = 0$. The last column represents the percentage of the adverse selection cost over that of the entire estimated spread. Accordingly, we reject the null hypothesis that the average adverse selection component

¹¹We repeat our full sample analyses with [Newey and West \(1987\)](#) standard errors. Our full sample results are robust to the potential heteroskedasticity and autocorrelation in the disturbance terms.

of the spread is zero for all cryptocurrencies in our sample period. XMR has the highest ratio for the adverse selection with 13%. Even though the BTC has the highest estimate for the adverse selection costs, the proportion of the spread attributable to adverse selection is one of the lowest in our sample with 7%. This is significantly smaller compared to findings of [Dimpfl \(2017\)](#)¹².

- INSERT TABLE 3 HERE -

Similarly, in Table 4, we document the adverse selection component of the spread using [Huang and Stoll \(1997\)](#) framework. The second and third columns respectively represent the estimated coefficient for the spread ($\hat{\nu}$) and its standard error. The third and fourth columns represent the estimate for the spread attributable to adverse selection and inventory holding costs ($ASC^{HS2} = \nu * \lambda$) and the corresponding standard errors. The last column gives the percentage of the spread that is attributed to adverse selection and inventory holding costs (λ). Both the estimated measures for adverse selection and order processing costs and the proportion of spread attributable to adverse selection are almost identical with the [Glosten and Harris \(1988\)](#) model. The adverse selection costs are significantly different from zero for all cryptocurrencies.

- INSERT TABLE 4 HERE -

Moreover, in Table 5, we provide results for the adverse selection component of the spread estimated using [Madhavan et al. \(1997\)](#) framework. The second and third columns in Table 5 presents the estimates for order processing costs and the corresponding standard errors, respectively. Moreover, in columns four and five we provide the estimate for adverse selection costs and the corresponding standard errors. Similar to the previous two models, we document that the adverse selection component of the spread is different than zero for all cryptocurrencies in our sample.

- INSERT TABLE 5 HERE -

Finally, from the two-way decomposition, we understand that the estimates for adverse selection costs ignore the role of inventory holding costs ([Huang and Stoll, 1997](#)). To account for the inventory

¹²[Dimpfl \(2017\)](#) shows that more than 50% of the spread is attributed to adverse selection for most of the Bitcoin markets. The differences in results might be attributed to different market characteristics, different currencies and/or different sample periods. In addition, [Dimpfl \(2017\)](#) ignores the inventory holding and order-processing costs in estimating the components of the spread.

holding costs, we use the three-way decomposition presented in equation (18). Table 6 present the results for the estimation of the three-way decomposition of the Huang and Stoll (1997). Columns two and three provide the estimate for the traded spread (ν) and the corresponding standard error. Columns four and five provide an estimate for the proportion of inventory costs in the estimated spread (β) and the corresponding standard errors. Similarly, columns six and seven present the estimate for the proportion of adverse selection costs in the estimated spread (α) and the corresponding standard error of the estimate. Finally, the last column gives the estimate for the serial correlation in the trade flow. Our results indicate that the proportion of inventory holding costs is highest for BTC which also has the highest estimate for the serial correlation in the trade flow. We observe that the proportion of adverse selection in the estimated spread is negative for all cryptocurrencies in our sample. This is because the estimated serial correlation in the trade flow (π) is less than 0.5. Even though this finding is in line with those of Huang and Stoll (1997) for U.S. equity markets, the negative spread has no meaning. Therefore, we exclude negative values for ASC, OPC and IC in our panel regressions in the following section as well.¹³

- INSERT TABLE 6 HERE -

To this end, we can argue that there is significant informed trading activity in cryptocurrency markets. The results of the two-way decomposition models consistently indicate that the adverse selection component of the spread is largest (smallest) for BTC (XRP) in dollar terms without normalizing for average trade price. In addition, with the three-way decomposition model of Huang and Stoll (1997), we observe that BTC has the highest serial correlation in the trade-flow and the highest inventory costs both in terms of proportion of the spread and in dollar terms. In the next part of our analysis, we test whether private information affects future returns, volatility of returns, liquidity, and toxicity in cryptocurrency markets.

4.2 Panel Regressions

For our panel analysis, we estimate the adverse selection of the spreads for all cryptocurrencies and all sampling frequencies (1-min, 15-min, and 60-min). Let $ASC_{i,t}^k$ be the adverse selection

¹³Because of the negative spread values, we remove a significant number of observations from our sample for three-way decomposition of Huang and Stoll (1997). For example, Table 1 shows that the number of observations for the two-way decomposition models is around 87000 whereas the estimate for three-way decomposition is around 28,000.

component of the spread for cryptocurrency i in time interval t estimated using model k which can either be the two-way decompositions of Glosten and Harris (1988); Madhavan et al. (1997); Huang and Stoll (1997) or the three-way decomposition of Huang and Stoll (1997), that is, $k \in \{GH, MRR, HS2, HS3\}$.

To examine the relationship between informed trading and return volatility (as in Hypothesis 1), we first estimate equation (19) with currency- and time-fixed effects. The results are presented in Table 7 in which Panel A, B and C display the results of the estimation for time intervals set to 1-min, 15-mins, and 60-mins, respectively. For all of the time intervals, we reject the null hypothesis that an increase in adverse selection does not lead to a change in future return volatility. For the relatively long intervals (15-mins and 60-mins), we document that future return volatility increases with an increase in the adverse selection component of the spread. This result is robust across different methods for calculating the adverse selection measure. Our findings are in line with those by Frömmel et al. (2008) who document that an increase in informed trading activity increases the volatility of currency returns. For the shortest interval (1-min), we observe the opposite sign and conclude that an increase in the adverse selection costs reduces the return volatility, a finding similar to that of Rzaev and Ibikunle (2019) for the U.S stock market. Since we have different signs for different time intervals, we conduct a robustness check where we run equation (19) with absolute price change $|\Delta P|$ as the dependent variable. We observe that the impact of adverse selection costs is positive for all time intervals when $|\Delta P|$ is used as a dependent variable.¹⁴ Therefore, we conclude that the relationship between informed trading activity and volatility in cryptocurrency markets is similar to the previous findings documented for currency markets rather than stock markets. This particular result is interesting as the discussion surrounding which asset class cryptocurrencies belong to continues¹⁵ (whether it be considered a stock, commodity, or currency). Within a limited capacity, our results support the asset's consideration as a currency when compared to other listed assets.

- INSERT TABLE 7 HERE -

Secondly, we examine the relationship between adverse selection costs and the future liquidity

¹⁴We do not report the table containing the results of our robustness check due to space considerations. It is available upon request.

¹⁵<http://aswathdamodaran.blogspot.com/2017/10/the-bitcoin-boom-asset-currency.html>

of cryptocurrencies. We use Amihud illiquidity (*ILLIQ*) measure along with realised spread (*SPREAD*) and order-book slope (*SLOPE*) as inverse measures of liquidity, and order-book depth (*DEPTH*) as a direct measure of liquidity. We expect the liquidity to go down with the increase in transaction costs (OPC, ASC and IC). Hence, our second hypothesis suggests a reverse relationship between adverse selection costs and future liquidity. To test this second hypothesis, we estimate equations (20), (21), (22), and (23) with currency- and time-fixed effects. Table 8 provides the results for equation (20) which uses *ILLIQ* as a dependent variable. The results indicate that the illiquidity of cryptocurrencies increases with an increase in the adverse selection component of the spread. Similarly, we present the estimates for equation (21) in Table 9 which indicates that the realised spread (*SPREAD*) also increases with information asymmetry, confirming our expectations that transaction costs are increasing with the presence of informed trading. We also observe a positive relationship between adverse selection costs and future levels of order-book slope (*SLOPE*) in Table 10 which provides additional evidence towards a reverse relationship between information asymmetry and liquidity. Finally, we examine the role of information asymmetry in explaining the future levels of order-book depth (*DEPTH*). According to Table 11, informed trading levels do not have a significant impact on order-book depth, unlike other liquidity measures.

INSERT TABLES 8-9-10-11 HERE

Overall, our results confirm that information asymmetry increases the transaction costs in cryptocurrency markets and adverse selection costs are significant determinants of future *illiquidity* of the cryptocurrency markets as proxied by the Amihud illiquidity measure, and order-book slope. Finally, we document that increased adverse selection costs do not have a statistically significant impact on future levels of order-book depth.

Next, we test whether the impact of an increase in informed trading activity affects the bid- and ask-side of the market differently. Recent studies underline the different attributes of bid- and ask-side liquidity (Roşu, 2009; Brennan et al., 2012; Cenesizoglu and Grass, 2018), and the presence of asymmetric adverse selection on the bid- and ask-side of the order book (Gencay et al., 2018). To this end, we compute the bid- and ask-side measures for *SLOPE* and *DEPTH* for each cryptocurrency and for all of the time intervals. Tables 12 and 13 respectively present the results for our panel regressions where the dependent variables are bid- and ask-side slope, respectively. We observe that

an increase in informed trading activity increases the order-book slope for *both* bid- and ask-side, reducing the liquidity on both sides of the trade. The coefficients of the adverse selection measures are quite close for both bid- and ask-side slopes in terms of magnitude, for each of the time intervals. Similarly, we show in Tables 14 and 15 that increased adverse selection has no significant effect on the order-book depth for *both* buy- and sell-side of the trade, respectively. Therefore, we can argue that informed trading has a *symmetric* effect on different sides of the liquidity of the cryptocurrencies.

- INSERT TABLES 12-13-14-15 HERE -

Considering our third hypothesis, we check the relationship between informed trading activity and market toxicity. As stated earlier, order-flow (market) toxicity is defined as the adverse selection costs faced by investors who are not aware that they provide liquidity at a loss, and it is argued that market toxicity can be considered as a high-frequency equivalent of adverse selection risk (Rzayev and Ibikunle, 2019). Therefore, we expect a positive relationship between adverse selection costs and future levels of toxicity. We define market toxicity as the ratio of absolute trade imbalance to the total number of trades and the results of our panel regressions are displayed in Table 16. Accordingly, we have puzzling findings. For all of the alternative measures of adverse selection and different time intervals, we observe that informed trading activity reduces the future levels of market toxicity. Our findings suggest that an increase in adverse selection cost reduces the future levels of informed trading activity. Even though this finding is puzzling, a potential channel may be attributed to the price discovery of informed trades in cryptocurrency markets. We may argue that informed trading activity in cryptocurrency markets results in a greater price discovery compared to other financial markets. Transaction costs are quickly adjusted in the presence of informed trading activity so that an increase in transaction costs due to adverse selection risks might make the (relatively short-term) trading based on private information less profitable, thus driving informed traders away from the market thereby resulting in lower levels of future toxicity¹⁶.

- INSERT TABLE 16 HERE -

¹⁶We check the impact of multicollinearity for all of the results presented above. The univariate relationship between ASC's and future risk, liquidity and toxicity levels is consistent under a univariate setting which rules out the multicollinearity problem for multivariate analysis. We do not present the univariate results due to space limitations, but they are available upon request.

Finally, we examine the relationship between informed trading and future returns. From the market efficiency perspective, we expect to see a negative association (Rzayev and Ibikunle, 2019), whereas, from the cost perspective, a positive association is expected (Biais et al., 2020). Chordia et al. (2002, 2008) show that informed trading activity proxied by lag order imbalances can be a significant predictor of short-horizon returns as investors gradually react to new information. In addition, Rzayev and Ibikunle (2019) indicates that informed order flow predicts the stock returns in one-second frequency. However, for relatively long time intervals such as one-minute frequency, informed trading does not predict future returns. Rzayev and Ibikunle (2019) attribute the lack of statistical significance to the price impact of high-frequency traders (HFT) in equity markets. They indicate that increased HFT volume helps increase the speed of information incorporation into stock prices and leads to the elimination of arbitrage opportunities. Consequently, we present our results for our adverse selection measures ASC_{GH} , ASC_{MRR} , ASC_{HS2} and ASC_{HS3} respectively in Tables 17, 18, 19 and 20. Our results indicate that an increase in the adverse selection of the spread is a significant and negative predictor of market efficiency for relatively long time intervals. This finding is consistent across all different measures of adverse selection components with two-way decomposition. On the contrary, we document that informed trading activity, proxied by the three-way decomposition of Huang and Stoll (1997) is not a significant predictor of future returns. However, it is important to note that our sample for the three-way decomposition of the spread is quite small. Since the serial correlations in the order flow are less than 0.5 for most of the time intervals, we frequently observe negative values for ASC_{HS3} and exclude them from our sample. Overall, the evidence suggested by the two-way decompositions of the spread asserts that informed trading activity proxied by the adverse selection component of the spread is a significant predictor of future cryptocurrency returns. We document that increases in the adverse selection costs reduce the future returns (at a 60-minute frequency). Consequently, we can argue that in cryptocurrency markets, transaction costs are adjusted so that the trading based on private information becomes less profitable within an hour¹⁷.

¹⁷Concerning Section 4.2, since the dependent variable in each panel regression shows up as an explanatory variable in all other panel regressions, we also design a vector auto-regression (VAR) setup and estimate of this VAR model for each cryptocurrency accordingly. Results are found to be qualitatively similar. We do not present the results here to save space, however, they can be obtained from the authors upon request and will also be made available as supplementary material to this paper.

5 Conclusion

Considering the substantial growth of cryptocurrency trading in recent years, an interesting question surrounds as to whether there are investors who possess superior information, which provides substantial benefit when assessing the fundamental values of these new instruments. Since all major cryptocurrencies are technological developments that serve a specific purpose, a group of investors could better assess net transactional benefits, and therefore, the fundamental value of associated products. To that end, investors who lack sufficient knowledge about cryptocurrencies face a significant adverse selection problem when trading with informed traders in secondary markets. Another source for a potential adverse selection problem in secondary cryptocurrency markets can be attributed to the informational differences between retail and institutional investors. Prior evidence suggests that institutional investors have substantial informational advantages, mainly due to their superior ability to monitor the conditions and process information at lower costs in traditional markets (Seppi, 1992; Hessel and Norman, 1992; Lang and McNichols, 1997). Even though institutional investors are becoming increasingly prevalent in cryptocurrency markets, the literature is silent on the impact of adverse selection in secondary cryptocurrency markets. In this paper, we aim to fill this gap by focusing on the role of information asymmetry in determining future volatility, liquidity, market toxicity, and return levels. Specifically, we focus on the most traded cryptocurrencies based on the Bitfinex exchange during a period when the exchange was the leading platform for secondary cryptocurrency trading activity. Using a high-frequency U.S. dollar-denominated order book and trade data, we initially show a statistically significant adverse selection component of the effective spread for all cryptocurrencies in our sample. Our results also suggest that, on average, 10% of the effective spread is attributable to adverse selection risks, indicating that the role of information asymmetry is not only statistically significant but also economically significant.

We further document that adverse selection costs significantly predict intraday volatility, liquidity, market toxicity, and returns. Specifically using intraday panel regressions, we show that, on average, future volatility levels increase with the increases in adverse selection costs in secondary cryptocurrency markets. We also document that informed traders demand liquidity in cryptocurrency

markets. In particular, our results suggest that increases in adverse selection costs are associated with overall decreases in future liquidity levels, as evidenced by positive predictive relationships with inverse liquidity measures such as realized spread, Amihud illiquidity measure, and order book slope. We further document that the predictive relationship between adverse selection costs and future liquidity is symmetric on both the bid- and the ask side of the limit order book. The impact of adverse selection costs on future levels of volatility and liquidity is in line with the previous findings documented for the currency markets (Payne, 2003; Frömmel et al., 2008); rather than recent evidence on stock markets (Rzayev and Ibikunle, 2019). These results may, therefore, have the potential to advance the discussions on which asset class cryptocurrencies belong, with evidence suggesting that it is best described as a currency, albeit within a limited capacity.

Finally, we document a negative predictive relationship between adverse selection costs and future market toxicity levels. This finding suggests that increases in transaction costs due to information asymmetry can lead to higher price discovery and drive investors with private information away from the market as trading based on private information becomes less profitable. This would lead to lower levels of market toxicity in the future. In line with this result, we also observe a significant and negative predictive relationship between adverse selection costs and future returns in 60-minute intervals. From the transaction cost perspective, the direction of the predictive relationship between adverse selection costs and future returns does not support the expectations of (Biais et al., 2020). However, the transaction costs in the framework of Biais et al. (2020) are related to the proxies indicative of the easiness of transactions with cryptocurrencies. Therefore, we argue that the increase in transaction costs due to adverse selection risk can be negligible compared to the significant events determining the future of cryptocurrency trading. On the other hand, from a market efficiency perspective, the inverse relationship between informed trading activity and future returns indicates that cryptocurrency markets are efficient at relatively short horizons Rzayev and Ibikunle (2019). Consequently, we can argue that transaction costs are adjusted in cryptocurrency markets so that trading based on private information becomes less profitable within an hour.

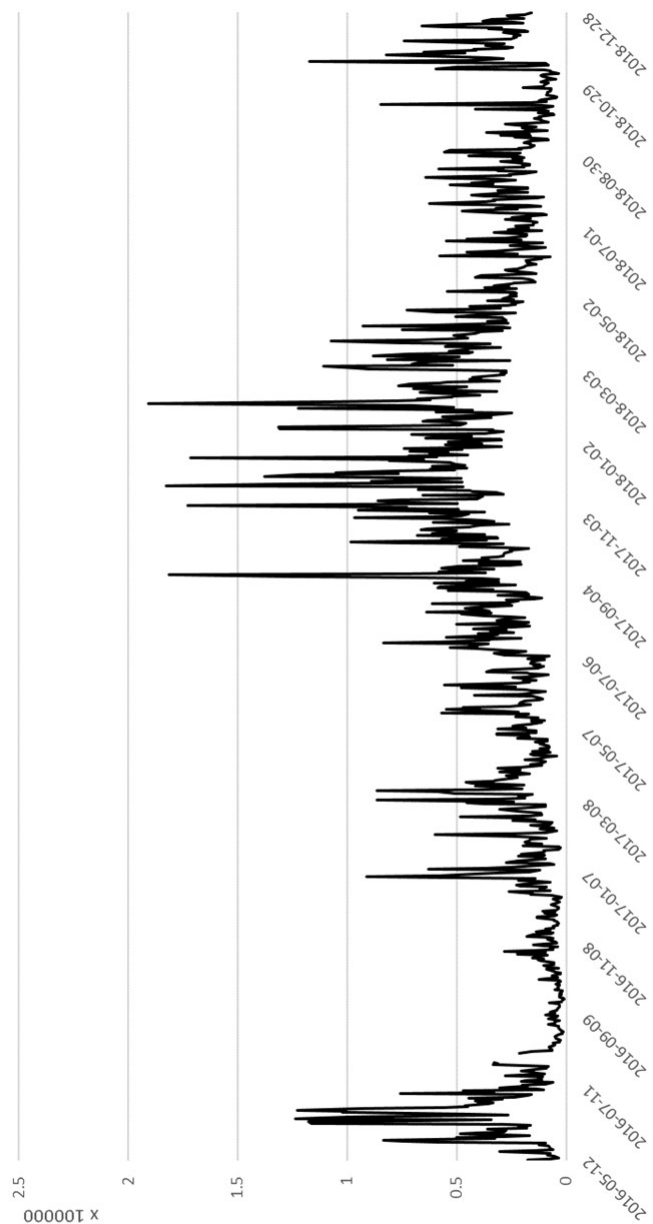
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Figure 1: Bitfinex Trading Volumes



Note: The above figure we see the trading volumes on the Bitfinex exchange. Data was obtained from Bitcoinity and are available [here](#).

Table 1: This table presents descriptive statistics for our variables calculated for 60 minute frequency sampling. For our panel analysis, we restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse.

Variable	Observations	Mean	Std. Dev.	Minimum	Maximum
$TV_{i,t}$	90,334	9.26	2.84	1.47	18.07
$BSI_{i,t}$	90,334	198.90	338.48	0.00	12842.00
$MT_{i,t}$	90,334	0.19	0.16	0.00	1.00
$R_{i,t}$	91,661	0.00	0.02	-0.33	0.37
$ILLIQ_{i,t}$	90,322	0.00	0.00	0.00	0.03
$\sigma_{i,t}^R$	90,334	0.01	0.01	0.00	0.25
$ASC_{i,t}^{GH}$	87,024	0.05	0.08	0.00	1.10
$ASC_{i,t}^{HS2}$	87,160	0.05	0.08	0.00	1.10
$ASC_{i,t}^{HS3}$	28,244	0.03	0.06	0.00	1.08
$ASC_{i,t}^{MRR}$	87,161	0.05	0.08	0.00	1.10
$OPC_{i,t}^{GH}$	87,024	0.37	0.83	0.00	31.79
$OPC_{i,t}^{HS2}$	87,160	0.37	0.82	0.00	31.72
$OPC_{i,t}^{HS3}$	28,244	0.30	0.66	0.00	18.07
$OPC_{i,t}^{MRR}$	87,161	0.37	0.82	0.00	31.72
$IC_{i,t}^{HS3}$	28,244	0.03	0.06	0.00	0.69
$SPREAD_{i,t}$	88,632	0.47	2.58	0.00	382.82
$DEPTH_{i,t}$	90,665	2.24	7.80	-0.09	931.13
$SLOPE_{i,t}$	88,634	0.02	0.08	-1.81	1.20
$SLOPE_{i,t}^B$	88,634	0.03	0.31	0.00	67.87
$SLOPE_{i,t}^A$	90,665	0.03	0.36	-0.05	84.77
$DEPTH_{i,t}^B$	90,665	1.15	4.18	0.00	321.91
$DEPTH_{i,t}^A$	90,665	1.08	5.25	-4.44	927.30

$TV_{i,t}$ is the logarithm of the U.S. dollar denominated total volume of trades for cryptocurrency i in time interval t . Buy-Sell imbalance is the absolute difference between buyer- and seller-initiated trades in time interval t for cryptocurrency i , i.e., $BSI_{i,t} = |B_{i,t} - S_{i,t}|$ where $B_{i,t}$ ($S_{i,t}$) is the number of buyer-initiated (seller-initiated) trades in the time interval t for cryptocurrency i . Market toxicity is defined as $MT_{i,t} = \frac{BSI_{i,t}}{(B_{i,t} + S_{i,t})}$. Return is defined as $R_{i,t} = \log(P_{i,t}/P_{i,t-1})$ where $P_{i,t}$ is the last price in time interval t for cryptocurrency i . $ILLIQ_{i,t}$ is a measure of cryptocurrency illiquidity for time interval t with the ratio of absolute stock return to the total trading volume defined as $ILLIQ_{i,t} = \frac{|R_{i,t}|}{TV_{i,t}}$. $\sigma_{i,t}^R$ is defined as the standard deviation of returns of the transactions in interval t for cryptocurrency i . $ASC_{GH}, ASC_{HS2}, ASC_{MRR}$ are three measures for the adverse selection component of the spread for all cryptocurrencies in our sample for all of the selected frequencies computed using the seminal works of [Glosten and Harris \(1988\)](#), [Huang and Stoll \(1997\)](#), and [Madhavan et al. \(1997\)](#), respectively. ASC_{HS3} denotes the adverse selection costs computed with the three-way decomposition of the [Huang and Stoll \(1997\)](#) framework. $OPC_{i,t}^{GH}, OPC_{i,t}^{HS2}, OPC_{i,t}^{HS3}, OPC_{i,t}^{MRR}$ are the transitory components of price changes related with the order processing costs (OPC) defined in the related paper as above. $IC_{i,t}^{HS3}$ is the inventory holding costs (IC) (in the case of [Huang and Stoll \(1997\)](#)) component of the price changes. $SPREAD_{i,t}$ is the average value of the difference between the best ask and bid price at each time tick in the time interval t for cryptocurrency i . $DEPTH_{i,t}$ is the average of the average volume waiting at the best bid and ask levels at each time tick in the time interval t for cryptocurrency i . $SLOPE_{i,t}$ denotes the marginal liquidity cost as in [Amaya et al. \(2018\)](#). $SLOPE_{i,t}^B$ ($SLOPE_{i,t}^A$) is the order-book slope computed from the bid-side (ask-side) of the order book. $DEPTH_{i,t}^B$ ($DEPTH_{i,t}^A$) is the average volume waiting at the best bid (ask) level at each time tick in the time interval t for cryptocurrency i .

Table 2: This table presents the pairwise correlations for our variables calculated over a time interval of 60 minutes. For our panel analysis, we restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse. †, ‡, * indicate statistical significance at 1%, 5% and 10% level, respectively.

$BSI_{i,t}$	$TV_{i,t}$	$BSI_{i,t}$	$MT_{i,t}$	$R_{i,t}$	$ILLIQ_{i,t}$	$\sigma_{i,t}^R$	$ASC_{i,t}^{GH}$	$ASC_{i,t}^{HS2}$	$ASC_{i,t}^{HS3}$	$ASC_{i,t}^{MRR}$	$OPC_{i,t}^{GH}$	$OPC_{i,t}^{HS2}$	$OPC_{i,t}^{HS3}$	$IC_{i,t}^{HS3}$	$SPREAD_{i,t}$	$DEPTH_{i,t}$
	0.16†															
$MT_{i,t}$	-0.15†	0.26†														
$R_{i,t}$	0.01	-0.03†	-0.04†													
$ILLIQ_{i,t}$	0.00	0.23†	0.04†	0.06†												
$\sigma_{i,t}^R$	0.34†	0.27†	-0.10†	0.03†	0.71†											
$ASC_{i,t}^{GH}$	-0.53†	0.10†	-0.09†	-0.00	0.14†	-0.01*										
$ASC_{i,t}^{HS2}$	-0.53†	0.10†	-0.09†	-0.01	0.14†	-0.01*	1.00†									
$ASC_{i,t}^{HS3}$	-0.50†	-0.00	-0.02‡	-0.01	0.14†	-0.01	0.75†	0.75†								
$ASC_{i,t}^{MRR}$	-0.53†	0.10†	-0.09†	-0.01	0.14†	-0.01*	1.00†	1.00†	0.75†							
$OPC_{i,t}^{GH}$	-0.30†	0.24†	-0.10†	-0.01*	0.21†	0.13†	0.68†	0.69†	0.56†	0.69†						
$OPC_{i,t}^{HS2}$	-0.30†	0.24†	-0.10†	-0.01*	0.22†	0.14†	0.68†	0.68†	0.56†	0.68†	1.00†					
$OPC_{i,t}^{HS3}$	-0.35†	0.20†	-0.09†	-0.02	0.22†	0.11†	0.72†	0.72†	0.54†	0.72†	1.00†	1.00†				
$IC_{i,t}^{HS3}$	-0.50†	0.03†	-0.06†	-0.01	0.15†	-0.00	0.89†	0.89†	0.38†	0.89†	0.63†	0.62†	0.62†			
$SPREAD_{i,t}$	-0.14†	0.04†	-0.03†	-0.00	0.05†	0.01†	0.23†	0.23†	0.23†	0.23†	0.23†	0.23†	0.29†	0.26†		
$DEPTH_{i,t}$	0.44†	-0.01†	-0.03†	0.00	-0.08†	0.03†	-0.18†	-0.18†	-0.12†	-0.18†	-0.13†	-0.13†	-0.11†	-0.13†	-0.05†	
$SLOPE_{i,t}$	-0.20†	0.27†	-0.11†	-0.00	0.04 †	-0.02†	0.64†	0.64†	0.38†	0.64†	0.67†	0.67†	0.67†	0.54†	0.20†	-0.08†

Table 3: Spread decomposition-(Glosten and Harris, 1988): This table presents the adverse selection components of the spread estimated using the Glosten and Harris (1988) framework. We estimate the equation $\Delta P = z_0 Q_t + z_1 Q_t V_t + c_0(\Delta Q_t) + c_1(\Delta Q_t V_t) + \epsilon_t$ using the limit order book of Bitfinex for all cryptocurrencies in our sample with ordinary least squares. The component of the spread that is attributed to adverse selection is given by $ASC_t^{GH} = 2 * (\hat{z}_0 + \hat{z}_1 V_t)$, and the component of the spread attributed to order-processing costs is given by $(OPC_t^{GH} = 2 * (\hat{c}_0 + \hat{c}_1 V_t))$. The second column is designated for the mean of the transitory component OPC_t^{GH} throughout the sample period. Third column provides p-value for the test with the null hypothesis that transitory component is equal to zero. The fourth column and fifth columns are designated for the mean of the adverse selection component of the spread (ASC_t^{GH}) and the corresponding p-value for the test with the null hypothesis that adverse selection component is equal to zero. The last column gives ratio of adverse selection costs overall traded spread ($ASC_t/(OPC_t + ASC_t)$).

Currency	Transitory Comp. (OPC_t^{GH})		Adverse Selection Comp. (ASC_t^{GH})		% ASC
	Coefficient	p-value	Coefficient	p-value	
BCH	1.8121	(0.00)	0.1509	(0.00)	8%
BTC	3.0157	(0.00)	0.2172	(0.00)	7%
DSH	1.0203	(0.00)	0.1268	(0.00)	11%
EOS	0.0090	(0.00)	0.0010	(0.00)	10%
ETC	0.0293	(0.00)	0.0033	(0.00)	10%
ETH	0.2702	(0.00)	0.0272	(0.00)	9%
IOT	0.0037	(0.00)	0.0003	(0.00)	7%
LTC	0.1490	(0.00)	0.0142	(0.00)	9%
OMG	0.0223	(0.00)	0.0026	(0.00)	10%
XMR	0.3777	(0.00)	0.0547	(0.00)	13%
XRP	0.0013	(0.00)	0.0001	(0.00)	7%
ZEC	0.6813	(0.00)	0.0807	(0.00)	11%

Table 4: Two-way spread decomposition - (Huang and Stoll, 1997): This table presents the adverse selection component of the spread estimated using the two-way decomposition of the Huang and Stoll (1997) framework. We estimate the equation $\Delta P_t = \lambda \frac{\nu}{2} Q_{t-1} + \frac{\nu}{2} (\Delta Q_t) + \epsilon_t$ using the limit order book of Bitfinex for all cryptocurrencies in our sample. The second column presents the traded spread (ν). The third column corresponds to the ordinary least squares standard error of the estimates of traded spread. The fourth and fifth columns present the estimates for the component of the spread attributed to the adverse selection (ASC) and inventory costs (IC) ($\lambda * \frac{\nu}{2}$) and corresponding ordinary least squares standard errors, respectively. The last column represents the proportion of the spread that is attributed to the adverse selection and inventory costs (λ).

Currency	Traded Spread (ν)		$ASC^{HS} + IC^{HS} (\lambda * \nu)$		λ
	Coefficient	Standard Error	Coefficient	Standard Error	
BCH	1.9602	(0.0012)	0.1510	(0.0011)	8%
BTC	3.2237	(0.0013)	0.2176	(0.0014)	7%
DSH	1.1445	(0.0011)	0.1269	(0.0009)	11%
EOS	0.0100	(0.0000)	0.0010	(0.0000)	10%
ETC	0.0325	(0.0000)	0.0033	(0.0000)	10%
ETH	0.2968	(0.0002)	0.0272	(0.0002)	9%
IOT	0.0039	(0.0000)	0.0003	(0.0000)	7%
LTC	0.1630	(0.0001)	0.0142	(0.0001)	9%
OMG	0.0249	(0.0000)	0.0026	(0.0000)	10%
XMR	0.7610	(0.0007)	0.0808	(0.0006)	11%
XRP	0.0014	(0.0000)	0.0001	(0.0000)	7%
ZEC	0.7610	(0.0007)	0.0808	(0.0006)	11%

Table 5: Spread decomposition - (Madhavan et al., 1997): This table presents the adverse selection component of the spread estimated using the two-way decomposition of the Madhavan et al. (1997) framework. We estimate the equation $\Delta P_t = (\phi + \theta)Q_t - \phi Q_{t-1} + \epsilon_t$ using the limit order book of Bitfinex for all cryptocurrencies in our sample. The component of the spread that is attributed to adverse selection is given by $ASC^{MRR} = 2 * (\hat{\theta})$, and the component of the spread attributed to order-processing costs is given by $(OPC^{MRR} = 2 * (\hat{\phi}))$. The second column is designated for the mean of the transitory component OPC_t^{MRR} throughout the sample period. Third column provides ordinary least squares standard errors for the OPC estimate. The fourth column and fifth columns are designated for the mean of the adverse selection component of the spread (ASC^{MRR}) and the corresponding ordinary least squares standard error for the ASC estimate. The last column gives ratio of adverse selection costs overall traded spread ($ASC^{MRR}/(OPC^{MRR} + ASC^{MRR})$).

Currency	Order Processing Costs (OPC^{MRR})		Adverse Sel. Costs (ASC^{MRR})		% ASC
	Coefficient	Std. Error	Coefficient	Std. Error	
BCH	1.8091	(0.0012)	0.1510	(0.0011)	8%
BTC	3.0061	(0.0013)	0.2176	(0.0014)	7%
DSH	1.0176	(0.0011)	0.1269	(0.0009)	11%
EOS	0.0090	(0.0000)	0.0010	(0.0000)	10%
ETC	0.0292	(0.0000)	0.0033	(0.0000)	10%
ETH	0.2696	(0.0002)	0.0272	(0.0002)	9%
IOT	0.0037	(0.0000)	0.0003	(0.0000)	7%
LTC	0.1488	(0.0001)	0.0142	(0.0001)	9%
OMG	0.0223	(0.0000)	0.0026	(0.0000)	10%
XMR	0.3770	(0.0005)	0.0547	(0.0005)	13%
XRP	0.0013	(0.0000)	0.0001	(0.0000)	7%
ZEC	0.6801	(0.0007)	0.0808	(0.0006)	11%

Table 6: Three-way spread decomposition - (Huang and Stoll, 1997): This table presents the adverse selection component of the spread estimated using the three-way decomposition of the Huang and Stoll (1997) framework. We estimate the equation $\Delta P_t = \frac{\nu}{2}Q_t + (\alpha + \beta - 1)\frac{\nu}{2}Q_{t-1} - \alpha\frac{\nu}{2}(1 - 2\pi)Q_{t-2} + \epsilon_t$ using the limit order book of Bitfinex for all cryptocurrencies in our sample. The second column presents the traded spread (ν). The third column corresponds to the ordinary least squares standard error of the estimates of traded spread. The fourth and fifth columns present the estimates for the proportion of spread due to the inventory holding costs ($\hat{\beta}$) and the corresponding ordinary least squares standard error. The sixth and seventh columns present the estimates for the proportion of the spread attributed to the adverse selection costs ($\hat{\alpha}$) and corresponding ordinary least squares standard error, respectively. The last column represents the serial correlation in the trade-flow (π).

Currency	Traded Spread (ν)				(%IC = β)				(%ASC = α)			
	Coefficient	Standard Error	Coefficient	Standard Error	Coefficient	Standard Error	Coefficient	Standard Error	Coefficient	Standard Error	Coefficient	π
BCH	1.9433	(0.0012)	0.1327	(0.0013)	-0.0828	(0.0022)	-0.0828	(0.0022)	-0.0828	(0.0022)	-0.0828	0.2324
BTC	3.1901	(0.0013)	0.1589	(0.0014)	-0.1180	(0.0031)	-0.1180	(0.0031)	-0.1180	(0.0031)	-0.1180	0.2880
DSH	1.1413	(0.0011)	0.1256	(0.0012)	-0.0249	(0.0018)	-0.0249	(0.0018)	-0.0249	(0.0018)	-0.0249	0.1930
EOS	0.0099	(0.0000)	0.1342	(0.0000)	-0.0554	(0.0000)	-0.0554	(0.0000)	-0.0554	(0.0000)	-0.0554	0.2103
ETC	0.0323	(0.0000)	0.1239	(0.0000)	-0.0377	(0.0000)	-0.0377	(0.0000)	-0.0377	(0.0000)	-0.0377	0.2057
ETH	0.2938	(0.0002)	0.1539	(0.0002)	-0.0901	(0.0003)	-0.0901	(0.0003)	-0.0901	(0.0003)	-0.0901	0.2420
IOT	0.0039	(0.0000)	0.1136	(0.0000)	-0.0771	(0.0000)	-0.0771	(0.0000)	-0.0771	(0.0000)	-0.0771	0.2137
LTC	0.1615	(0.0001)	0.1487	(0.0001)	-0.0898	(0.0002)	-0.0898	(0.0002)	-0.0898	(0.0002)	-0.0898	0.2378
OMG	0.0248	(0.0000)	0.1100	(0.0000)	-0.0117	(0.0000)	-0.0117	(0.0000)	-0.0117	(0.0000)	-0.0117	0.1871
XMR	0.4309	(0.0005)	0.1376	(0.0006)	-0.0181	(0.0009)	-0.0181	(0.0009)	-0.0181	(0.0009)	-0.0181	0.1951
XRP	0.0014	(0.0000)	0.1386	(0.0000)	-0.1009	(0.0000)	-0.1009	(0.0000)	-0.1009	(0.0000)	-0.1009	0.2534
ZEC	0.7597	(0.0007)	0.1153	(0.0008)	-0.0152	(0.0012)	-0.0152	(0.0012)	-0.0152	(0.0012)	-0.0152	0.1950

Table 7: This table presents the results of the following regression: $\sigma_{i,t}^R = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \epsilon_{i,t}$. We restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse. Panel A, B, C present the results of the estimation for time intervals set to 1 minute, 15 minutes and 60 minutes, respectively. Respective columns in each panel are for the regressions where the components of the spread ASC^k , OPC^k are obtained by [Glosten and Harris \(1988\)](#) ($k = GH$), two-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS2 - way$), [Madhavan et al. \(1997\)](#) ($k = MRR$), and three-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS3 - way$). Inventory costs (IC) are obtained only by the three-way decomposition of [Huang and Stoll \(1997\)](#). We require at least 30 trades within the time interval t , to estimate the adverse selection component in that interval. Absolute value of the t -statistics are in parenthesis. Hypothesis 1 asserts a relationship $\gamma_6 \neq 0$. ***, **, * indicates statistical significance at 1%, 5% and 10%, respectively.

	Panel A: 1 min				Panel B: 15 min				Panel C: 60 min			
	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way
$\sigma_{i,t}^R$	0.48698 (372.442)***	0.48077 (349.171)***	0.48058 (348.493)***	0.42513 (117.331)***	0.50480 (253.318)***	0.50453 (253.637)***	0.50458 (253.673)***	0.50203 (124.736)***	0.40922 (108.081)***	0.40929 (108.164)***	0.40929 (108.165)***	0.38369 (55.291)***
$SPREAD_{i,t-1}$	0.00000 (1.724)*	0.00000 (1.811)*	0.00000 (1.810)*	0.00000 (2.039)**	0.00000 (1.421)	0.00000 (1.460)	0.00000 (1.459)	-0.00000 (0.289)	0.00002 (2.881)***	0.00002 (2.860)***	0.00002 (2.860)***	0.00003 (1.647)*
$TV_{i,t-1}$	0.00009 (36.120)***	0.00009 (38.036)***	0.00009 (37.979)***	0.00011 (16.643)***	0.00029 (46.369)***	0.00029 (46.619)***	0.00029 (46.578)***	0.00030 (24.638)***	0.00081 (39.400)***	0.00081 (39.366)***	0.00081 (39.367)***	0.00094 (23.829)***
$BSI_{i,t-1}$	-0.00000 (11.608)***	-0.00000 (10.970)***	-0.00000 (11.060)***	-0.00000 (6.516)***	0.00000 (24.149)***	0.00000 (24.109)***	0.00000 (24.102)***	0.00000 (13.652)***	0.00000 (5.906)***	0.00000 (5.947)***	0.00000 (5.947)***	0.00000 (3.878)***
$MT_{i,t-1}$	-0.00014 (12.917)***	-0.00013 (12.960)***	-0.00013 (12.711)***	-0.00006 (2.115)**	-0.00072 (22.225)***	-0.00072 (22.367)***	-0.00072 (22.365)***	-0.00075 (12.318)***	-0.00102 (8.358)***	-0.00102 (8.340)***	-0.00102 (8.342)***	-0.00112 (4.942)***
$ASC_{i,t-1}$	-0.00002 (2.655)**	-0.00002 (2.541)**	-0.00002 (2.529)**	-0.00002 (17.011)***	0.00073 (9.614)***	0.00073 (9.629)***	0.00073 (9.632)***	0.00036 (2.454)**	0.00479 (14.027)***	0.00486 (14.225)***	0.00486 (14.225)***	0.00121 (1.748)*
$OPC_{i,t-1}$	0.00002 (15.932)***	0.00002 (16.982)***	0.00002 (16.982)***	0.00002 (17.011)***	0.00019 (25.041)***	0.00019 (25.168)***	0.00019 (25.173)***	0.00019 (2.501)**	0.00036 (12.709)***	0.00036 (12.708)***	0.00036 (12.708)***	0.00048 (6.854)***
$ASC_{i,t-1}$	0.00062 (45.731)***	0.00061 (44.950)***	0.00061 (44.816)***	0.00057 (15.562)***	-0.00022 (5.419)***	-0.00023 (5.454)***	-0.00022 (5.420)***	-0.00032 (3.826)***	-0.00290 (18.036)***	-0.00289 (18.033)***	-0.00289 (18.033)***	(5.804)*** (-0.00357 (11.371)***
$IC_{i,t-1}$												
<i>Intercept</i>												
R^2	0.38	0.37	0.37	0.33	0.39	0.39	0.39	0.40	0.33	0.33	0.33	0.30
Currency and Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	488,796	494,244	493,641	74,528	254,801	255,657	255,660	65,729	83,808	83,941	83,942	27,231

Table 8: This table presents the results of the following regression: $ILLIQ_{i,t} = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \epsilon_{i,t}$. We restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse. Panels A, B, C present the results of the estimation for time intervals set to 1 minute, 15 minutes and 60 minutes, respectively. Respective columns in each panel are for the regressions where the components of the spread ASC^k , OPC^k are obtained by [Glosten and Harris \(1988\)](#) ($k = GH$), two-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS2 - way$), [Madhavan et al. \(1997\)](#) ($k = MRR$), and three-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS3 - way$). Inventory costs (IC) are obtained only by the three-way decomposition of [Huang and Stoll \(1997\)](#). We require at least 30 trades within the time interval t , to estimate the adverse selection component in that interval. Absolute value of the t -statistics are in parenthesis. Hypothesis 2 asserts a relationship $\gamma_6 > 0$. ***, ***, * indicates statistical significance at 1%, 5% and 10%, respectively.

	Panel A: 1 min				Panel B: 15 min				Panel C: 60 min			
	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way
$\sigma_{i,t-1}^R$	0.08964 (133.481)***	0.09774 (163.046)***	0.09767 (162.767)***	0.07919 (65.893)***	0.08141 (129.541)***	0.08144 (129.789)***	0.08145 (129.816)***	0.08643 (63.835)***	0.05953 (54.241)***	0.05951 (54.242)***	0.05951 (54.242)***	0.05501 (28.061)***
$SPREAD_{i,t-1}$	0.00000 (1.687)*	0.00000 (2.159)**	0.00000 (2.158)**	0.00000 (0.405)	0.00000 (2.453)**	0.00000 (2.459)**	0.00000 (2.487)**	-0.00000 (-0.444)	0.00001 (4.554)***	0.00001 (4.527)***	0.00001 (4.527)***	0.00001 (2.840)***
$TV_{i,t-1}$	0.00001 (10.248)***	0.00001 (10.564)***	0.00001 (10.402)***	0.00001 (6.174)**	0.00004 (20.233)***	0.00004 (20.111)***	0.00004 (20.073)***	0.00003 (7.702)***	0.00011 (17.957)***	0.00011 (18.079)***	0.00011 (18.081)***	0.00012 (11.255)***
$BSI_{i,t-1}$	0.00000 (2.797)**	0.00000 (0.597)	0.00000 (0.590)	-0.00000 (-1.387)	0.00000 (4.901)***	0.00000 (4.897)***	0.00000 (4.896)***	0.00000 (3.589)***	-0.00000 (-0.840)	-0.00000 (-0.897)	-0.00000 (-0.895)	-0.00000 (-0.220)
$MT_{i,t-1}$	-0.00008 (13.999)***	-0.00007 (14.753)***	-0.00007 (14.520)***	-0.00005 (5.398)	-0.00015 (14.957)***	-0.00015 (15.106)***	-0.00015 (15.120)***	-0.00016 (7.867)***	-0.00020 (5.563)***	-0.00019 (5.489)***	-0.00019 (5.495)***	-0.00022 (3.410)***
$ASC_{i,t-1}$	0.00007 (17.204)***	0.00007 (39.195)***	0.00007 (19.587)***	0.00002 (44.851)***	0.00062 (25.824)***	0.00062 (41.993)***	0.00062 (42.044)***	0.00062 (8.480)***	0.00197 (19.876)***	0.00201 (20.260)***	0.00201 (20.260)***	0.00056 (2.865)***
$OPC_{i,t-1}$	0.00002 (44.854)***	0.00002 (44.854)***	0.00002 (44.851)***	0.00002 (3.133)***	0.00010 (22.461)***	0.00010 (42.044)***	0.00010 (42.047)***	0.00010 (19.610)***	0.00014 (17.169)***	0.00014 (17.043)***	0.00014 (17.042)***	0.00019 (9.587)***
$ASC_{i,t-1}$	0.00053 (76.284)***	0.00053 (89.109)***	0.00053 (89.082)***	0.00052 (4.568)***	0.00044 (33.504)***	0.00044 (33.828)***	0.00044 (33.859)***	0.00044 (15.833)***	0.00017 (3.746)***	0.00016 (3.528)***	0.00016 (3.528)***	0.00020 (9.037)***
$OPC_{i,t-1}$	0.00053 (76.284)***	0.00053 (89.109)***	0.00053 (89.082)***	0.00052 (4.568)***	0.00044 (33.504)***	0.00044 (33.828)***	0.00044 (33.859)***	0.00044 (15.833)***	0.00017 (3.746)***	0.00016 (3.528)***	0.00016 (3.528)***	0.00007 (0.777)
$IC_{i,t-1}$	0.00053 (76.284)***	0.00053 (89.109)***	0.00053 (89.082)***	0.00052 (4.568)***	0.00044 (33.504)***	0.00044 (33.828)***	0.00044 (33.859)***	0.00044 (15.833)***	0.00017 (3.746)***	0.00016 (3.528)***	0.00016 (3.528)***	0.00007 (0.777)
$Intercept$	0.00053 (76.284)***	0.00053 (89.109)***	0.00053 (89.082)***	0.00052 (4.568)***	0.00044 (33.504)***	0.00044 (33.828)***	0.00044 (33.859)***	0.00044 (15.833)***	0.00017 (3.746)***	0.00016 (3.528)***	0.00016 (3.528)***	0.00007 (0.777)
R^2	0.10 Yes	0.13 Yes	0.13 Yes	0.18 Yes	0.21 Yes	0.21 Yes	0.21 Yes	0.21 Yes	0.15 Yes	0.15 Yes	0.15 Yes	0.14 Yes
Currency and Time FE	488,796	494,244	493,641	74,528	254,801	255,657	255,660	65,729	83,808	83,941	83,942	27,231
N												

Table 9: This table presents the results of the following regression: $SPREAD_{i,t} = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \epsilon_{i,t}$. We restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse. Panels A, B, C present the results of the estimation for time intervals set to 1 minute, 15 minutes and 60 minutes, respectively. Respective columns in each panel are for the regressions where the components of the spread ASC^k, OPC^k are obtained by [Glosten and Harris \(1988\)](#) ($k = GH$), two-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS2 - way$), [Madhavan et al. \(1997\)](#) ($k = MRR$), and three-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS3 - way$). Inventory costs (IC) are obtained only by the three-way decomposition of [Huang and Stoll \(1997\)](#). We require at least 30 trades within the time interval t , to estimate the adverse selection component in that interval. Absolute value of the t -statistics are in parenthesis. Hypothesis 2 asserts a relationship $\gamma_6 > 0$. ***, * indicates statistical significance at 1% , 5% and 10%, respectively.

	Panel A: 1 min				Panel B: 15 min				Panel C: 60 min			
	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way
$\sigma_{i,t-1}^R$	-7.14630 (1.418)	-9.89233 (2.084)**	-9.69756 (2.041)**	-10.23819 (0.992)	-6.23575 (3.446)***	-6.50632 (3.601)***	-6.50551 (3.600)***	-8.01543 (6.080)***	-0.65937 (0.328)	-0.72521 (0.362)	-0.72510 (0.361)	-4.21980 (1.105)
$SPREAD_{i,t-1}$	0.04029 (90.066)***	0.02879 (71.123)***	0.02879 (71.082)***	0.04116 (13.557)***	0.15638 (142.293)***	0.14924 (139.030)***	0.14924 (139.030)***	0.03622 (61.590)***	0.11915 (33.623)***	0.11911 (33.640)***	0.11911 (33.640)***	0.08353 (6.183)***
$TV_{i,t-1}$	-0.05542 (6.679)***	-0.05581 (7.505)***	-0.05617 (7.538)***	-0.05333 (3.249)***	0.01660 (2.996)***	0.01729 (3.131)***	0.01729 (3.130)***	0.00724 (1.840)*	0.03342 (3.067)***	0.03393 (3.121)***	0.03393 (3.121)***	0.03301 (1.535)
$BSI_{i,t-1}$	0.00078 (2.750)***	0.00067 (2.613)***	0.00067 (2.619)***	0.00016 (0.293)	-0.00004 (0.882)	-0.00004 (0.882)	-0.00004 (0.883)	0.00002 (0.591)	-0.00011 (3.210)***	-0.00011 (3.225)***	-0.00011 (3.225)***	-0.00011 (1.538)
$MT_{i,t-1}$	-0.29098 (8.257)***	-0.29426 (9.325)***	-0.29556 (9.345)***	-0.17769 (2.578)***	-0.07462 (2.659)***	-0.07908 (2.834)***	-0.07904 (2.832)***	-0.05212 (2.717)***	0.03663 (0.566)	0.03944 (0.612)	0.03948 (0.612)	0.04746 (0.384)
$ASC_{i,t-1}$	0.81441 (26.634)***				1.30285 (19.480)***				1.83279 (10.153)***			
$OPC_{i,t-1}$	0.20734 (43.123)***				0.37355 (55.201)***				0.40139 (27.131)***			
$ASC_{i,t-1}$		0.86141 (30.746)***				1.33118 (19.864)***				1.87330 (10.374)***		
$OPC_{i,t-1}$		0.21154 (48.258)***				0.37860 (55.710)***				0.40133 (27.038)***		
$ASC_{i,t-1}$			0.86298 (30.777)***				1.33102 (19.861)***				1.87330 (10.374)***	
$OPC_{i,t-1}$			0.21149 (48.215)***				0.37860 (55.709)***				0.40133 (27.039)***	
$ASC_{i,t-1}$				0.58330 (8.782)***				1.16307 (24.741)***				1.34870 (3.561)***
$OPC_{i,t-1}$				0.22079 (16.674)***				0.51667 (82.025)***				0.40821 (10.631)***
$IC_{i,t-1}$				0.97141 (11.174)***				1.57113 (26.502)***				1.86441 (4.278)***
$Intercept$	1.65253 (35.897)***	1.62457 (39.360)***	1.62499 (39.313)***	1.44865 (15.682)***	0.57804 (15.836)***	0.57946 (15.919)***	0.57951 (15.920)***	0.54301 (20.519)***	0.34862 (4.105)***	0.34129 (4.028)***	0.34129 (4.028)***	0.36344 (2.115)**
R^2	0.04 Yes	0.04 Yes	0.04 Yes	0.04 Yes	0.14 Yes	0.14 Yes	0.14 Yes	0.42 Yes	0.08 Yes	0.08 Yes	0.08 Yes	0.06 Yes
Currency and Time FE	672,571	681,234	680,260	107,413	267,512	268,488	268,486	68,732	84,020	84,153	84,154	27,257
N												

Table 10: This table presents the results of the following regression: $SLOPE_{i,t} = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \epsilon_{i,t}$. We restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse. Panels A, B, C present the results of the estimation for time intervals set to 1 minute, 15 minutes and 60 minutes, respectively. Respective columns in each panel are for the regressions where the components of the spread ASC^k , OPC^k are obtained by [Glosten and Harris \(1988\)](#) ($k = GH$), two-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS2 - way$), [Madhavan et al. \(1997\)](#) ($k = MRR$), and three-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS3 - way$). Inventory costs (IC) are obtained only by the three-way decomposition of [Huang and Stoll \(1997\)](#). We require at least 30 trades within the time interval t , to estimate the adverse selection component in that interval. Absolute value of the t -statistics are in parenthesis. Hypothesis 2 asserts a relationship $\gamma_6 > 0$. ***, **, * indicates statistical significance at 1% , 5% and 10%, respectively.

	Panel A: 1 min				Panel B: 15 min				Panel C: 60 min			
	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way
$\sigma_{i,t-1}^R$	-1.98957 (24.780)***	-2.20153 (26.475)***	-2.18742 (26.276)***	-2.56028 (14.365)***	-1.44026 (37.533)***	-1.44901 (37.886)***	-1.44884 (37.882)***	-1.48281 (21.709)***	-0.95672 (25.806)***	-0.95870 (25.880)***	-0.95870 (25.880)***	-0.69982 (14.527)***
$SPREAD_{i,t-1}$	0.00020 (29.113)***	0.00020 (29.347)***	0.00020 (29.319)***	0.00102 (26.163)***	0.00043 (18.593)***	0.00041 (17.841)***	0.00041 (17.841)***	0.00009 (2.838)***	0.00098 (15.020)***	0.00098 (15.014)***	0.00098 (15.014)***	0.00285 (16.744)***
$TV_{i,t-1}$	-0.00751 (56.803)***	-0.00717 (55.010)***	-0.00721 (55.223)***	-0.00563 (19.866)***	0.00099 (8.410)***	0.00103 (8.834)***	0.00103 (8.822)***	0.00113 (5.521)***	0.00145 (7.213)***	0.00146 (7.287)***	0.00146 (7.287)***	-0.00035 (1.289)
$BSI_{i,t-1}$	0.00015 (33.101)***	0.00014 (32.325)***	0.00015 (32.454)***	0.00011 (12.501)***	0.00003 (31.270)***	0.00003 (31.156)***	0.00003 (31.165)***	0.00002 (14.479)***	0.00001 (16.284)***	0.00001 (16.353)***	0.00001 (16.353)***	0.00001 (15.163)***
$MT_{i,t-1}$	-0.03037 (54.087)***	-0.02938 (53.152)***	-0.02951 (53.264)***	-0.02394 (20.121)***	-0.00960 (16.142)***	-0.00929 (15.733)***	-0.00931 (15.751)***	-0.00691 (6.948)***	-0.00226 (1.895)*	-0.00215 (1.805)*	-0.00215 (1.805)*	-0.00329 (2.106)**
$ASC_{i,t-1}$	0.08406 (172.538)***				0.20745 (146.278)***				0.32489 (97.486)***			
$OPC_{i,t-1}$	0.01759 (229.595)***				0.02728 (190.142)***				0.03150 (115.319)***			
$ASC_{i,t-1}$	0.08726 (177.787)***				0.21040 (148.323)***					0.32807 (98.390)***		
$OPC_{i,t-1}$	0.01772 (230.805)***				0.02723 (189.279)***					0.03136 (114.401)***		
$ASC_{i,t-1}$		0.08726 (177.639)***				0.21042 (148.336)***				0.32807 (98.391)***		
$OPC_{i,t-1}$		0.01772 (230.588)***				0.02723 (189.280)***				0.03136 (114.402)***		
$ASC_{i,t-1}$			0.05506 (48.028)***					0.10920 (44.834)***				0.10659 (22.304)***
$OPC_{i,t-1}$			0.01855 (81.225)***					0.03198 (97.989)***				0.03190 (65.831)***
$OPC_{i,t-1}$			0.07988 (53.250)***					0.19354 (63.011)***				0.26291 (47.803)***
$IC_{i,t-1}$	0.02671 (36.418)***	0.02508 (34.688)***	0.02521 (34.819)***	0.01150 (7.214)***	-0.02873 (37.117)***	-0.02916 (37.840)***	-0.02915 (37.836)***	-0.03249 (23.698)***	-0.04985 (31.800)***	-0.05015 (32.049)***	-0.05015 (32.049)***	-0.02832 (13.057)***
$Intercept$												
R^2	0.55 Yes	0.55 Yes	0.55 Yes	0.56 Yes	0.65 Yes	0.65 Yes	0.65 Yes	0.65 Yes	0.70 Yes	0.70 Yes	0.70 Yes	0.65 Yes
Currency and Time FE	672,587	681,249	680,275	107,416	267,513	268,489	268,487	68,732	84,022	84,155	84,156	27,259
N												

Table 11: This table presents the results of the following regression: $DEPTH_{i,t} = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \epsilon_{i,t}$. We restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse. Panels A, B, C present the results of the estimation for time intervals set to 1 minute, 15 minutes and 60 minutes, respectively. Respective columns in each panel are for the regressions where the components of the spread ASC^k , OPC^k are obtained by [Glosten and Harris \(1988\)](#) ($k = GH$), two-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS2 - way$), [Madhavan et al. \(1997\)](#) ($k = MRR$), and three-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS3 - way$). Inventory costs (IC) are obtained only by the three-way decomposition of [Huang and Stoll \(1997\)](#). We require at least 30 trades within the time interval t , to estimate the adverse selection component in that interval. Absolute value of the t -statistics are in parenthesis. Hypothesis 2 asserts a relationship $\gamma_6 < 0$. ***, * indicates statistical significance at 1% , 5% and 10%, respectively.

	Panel A: 1 min			Panel B: 15 min			Panel C: 60 min		
	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way	HS 3-way
$\sigma_{i,t-1}^R$	86.22129 (3.251)***	98.01745 (3.598)**	98.11543 (3.597)**	64.51725 (0.589)	27.94706 (4.068)***	27.99134 (4.070)**	28.01963 (4.075)**	32.10648 (3.169)**	9.64745 (1.939)
$SPREAD_{i,t-1}$	0.00019 (0.002)	0.00020 (0.087)	0.00020 (0.087)	0.00197 (0.083)	0.00012 (0.029)	0.00008 (0.019)	0.00008 (0.019)	0.00016 (0.035)	-0.00043 (0.052)
$TV_{i,t-1}$	0.31264 (7.160)***	0.29761 (6.974)**	0.29852 (6.981)**	0.45269 (2.600)**	0.15594 (7.411)***	0.15577 (7.410)**	0.15552 (7.399)**	0.17919 (5.924)**	0.06840 (2.538)*
$BSI_{i,t-1}$	-0.00754 (5.037)***	-0.00734 (5.003)**	-0.00731 (4.973)**	-0.00554 (0.987)	-0.00221 (12.470)***	-0.00222 (12.540)**	-0.00222 (12.530)**	-0.00256 (10.036)**	-0.00068 (8.167)***
$MT_{i,t-1}$	0.93103 (5.020)***	0.91154 (5.034)**	0.90560 (4.989)**	1.30560 (1.786)	1.66435 (15.621)***	1.67457 (15.765)**	1.67251 (15.742)**	1.68999 (11.463)**	1.77554 (11.094)***
$ASC_{i,t-1}$	0.03054 (0.190)				0.11330 (0.446)			0.28920 (0.646)	
$OPC_{i,t-1}$	-0.01558 (0.616)				0.02061 (0.802)			0.03663 (0.999)	
$ASC_{i,t-1}$	0.02222 (0.138)					0.12035 (0.472)		0.29729 (0.664)	0.29726 (0.664)
$OPC_{i,t-1}$	-0.01650 (0.657)					0.02113 (0.816)		0.03631 (0.987)	0.03629 (0.986)
$ASC_{i,t-1}$									
$OPC_{i,t-1}$									
$ASC_{i,t-1}$									
$OPC_{i,t-1}$									
$IC_{i,t-1}$									
$Intercept$	-1.68967 (6.976)***	-1.63367 (6.899)**	-1.63620 (6.899)**	-2.51826 (2.572)*	-1.25120 (9.026)***	-1.25400 (9.050)**	-1.25206 (9.036)**	-1.41618 (6.964)**	-0.78051 (3.716)**
R^2	0.03 Yes	0.03 Yes	0.03 Yes	0.01 Yes	0.26 Yes	0.26 Yes	0.26 Yes	0.41 Yes	0.43 Yes
Currency and Time FE	674,322	683,001	682,030	107,673	267,925	268,902	268,901	68,839	84,451
N									84,452
									27,357

Table 12: This table presents the results of the following regression: $SLOPE_{i,t}^B = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \epsilon_{i,t}$. We restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse. Panel A, B, C present the results of the estimation for time intervals set to 1 minute, 15 minutes and 60 minutes, respectively. Respective columns in each panel are for the regressions where the components of the spread ASC^k , OPC^k are obtained by [Glosten and Harris \(1988\)](#) ($k = GH$), two-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS2 - way$), [Madhavan et al. \(1997\)](#) ($k = MRR$), and three-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS3 - way$). Inventory costs (IC) are obtained only by the three-way decomposition of [Huang and Stoll \(1997\)](#). We require at least 30 trades within the time interval t , to estimate the adverse selection component in that interval. Absolute value of the t -statistics are in parenthesis. Hypothesis 2 asserts a relationship $\gamma_6 > 0$. ***, * indicates statistical significance at 1% , 5% and 10%, respectively.

	Panel A: 1 min				Panel B: 15 min				Panel C: 60 min			
	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way
$\sigma_{i,t-1}^R$	-2.47912 (12.704)***	-2.74319 (13.566)***	-2.72787 (13.474)***	-2.90544 (14.142)***	-1.39144 (4.138)***	-1.38521 (4.133)***	-1.38520 (4.133)***	-1.21207 (2.665)***	-0.86384 (3.681)***	-0.86934 (3.708)***	-0.86934 (3.708)***	-0.48268 (1.335)
$SPREAD_{i,t-1}$	0.00022 (13.038)***	0.00022 (13.099)***	0.00022 (13.094)***	0.00106 (23.410)***	0.00051 (2.499)**	0.00048 (2.422)**	0.00048 (2.422)**	0.00013 (0.639)	0.00128 (3.084)***	0.00127 (3.079)***	0.00127 (3.079)***	0.00378 (2.955)***
$TV_{i,t-1}$	-0.00793 (24.675)***	-0.00758 (23.918)***	-0.00763 (24.021)***	-0.00574 (17.584)***	-0.00033 (0.317)	-0.00031 (0.306)	-0.00031 (0.307)	-0.00048 (0.351)	0.00024 (0.185)	0.00029 (0.228)	0.00029 (0.228)	-0.00249 (1.221)
$BSI_{i,t-1}$	0.00014 (12.655)***	0.00013 (12.185)***	0.00013 (12.257)***	0.00010 (9.364)***	0.00003 (2.919)***	0.00003 (2.913)***	0.00003 (2.913)***	0.00001 (0.776)	0.00001 (2.091)**	0.00001 (2.103)**	0.00001 (2.103)**	0.00001 (1.880)*
$MT_{i,t-1}$	-0.03177 (23.284)***	-0.03047 (22.663)***	-0.03060 (22.710)***	-0.02286 (16.665)***	-0.00677 (1.298)	-0.00671 (1.296)	-0.00672 (1.297)	0.01349 (2.039)**	-0.00339 (0.448)	-0.00326 (0.433)	-0.00326 (0.433)	-0.00579 (0.493)
$ASC_{i,t-1}$	0.09610 (81.164)***	0.09610 (81.164)***	0.09610 (81.164)***	0.09610 (81.164)***	0.17623 (14.181)***	0.17623 (14.181)***	0.17623 (14.181)***	0.28399 (13.462)***	0.28399 (13.462)***	0.28399 (13.462)***	0.28399 (13.462)***	0.28399 (13.462)***
$OPC_{i,t-1}$	0.01976 (106.098)***	0.01976 (106.098)***	0.01976 (106.098)***	0.01976 (106.098)***	0.02833 (22.531)***	0.02833 (22.531)***	0.02833 (22.531)***	0.02833 (22.531)***	0.03429 (19.830)***	0.03429 (19.830)***	0.03429 (19.830)***	0.03429 (19.830)***
$ASC_{i,t-1}$	0.09938 (83.273)***	0.09938 (83.273)***	0.09938 (83.273)***	0.09938 (83.273)***	0.17491 (14.071)***	0.17491 (14.071)***	0.17491 (14.071)***	0.17491 (14.071)***	0.28974 (13.730)***	0.28974 (13.730)***	0.28974 (13.730)***	0.28974 (13.730)***
$OPC_{i,t-1}$	0.01996 (106.901)***	0.01996 (106.901)***	0.01996 (106.901)***	0.01996 (106.901)***	0.02831 (22.457)***	0.02831 (22.457)***	0.02831 (22.457)***	0.02831 (22.457)***	0.03406 (19.636)***	0.03406 (19.636)***	0.03406 (19.636)***	0.03406 (19.636)***
$ASC_{i,t-1}$	0.09938 (83.192)***	0.09938 (83.192)***	0.09938 (83.192)***	0.09938 (83.192)***	0.17492 (14.072)***	0.17492 (14.072)***	0.17492 (14.072)***	0.17492 (14.072)***	0.28974 (13.731)***	0.28974 (13.731)***	0.28974 (13.731)***	0.28974 (13.731)***
$OPC_{i,t-1}$	0.01995 (106.765)***	0.01995 (106.765)***	0.01995 (106.765)***	0.01995 (106.765)***	0.02831 (22.457)***	0.02831 (22.457)***	0.02831 (22.457)***	0.02831 (22.457)***	0.03406 (19.637)***	0.03406 (19.637)***	0.03406 (19.637)***	0.03406 (19.637)***
$ASC_{i,t-1}$	0.05532 (41.868)***	0.05532 (41.868)***	0.05532 (41.868)***	0.05532 (41.868)***	0.08004 (4.936)***	0.08004 (4.936)***	0.08004 (4.936)***	0.08004 (4.936)***	0.08004 (4.936)***	0.08004 (4.936)***	0.08004 (4.936)***	0.08004 (4.936)***
$OPC_{i,t-1}$	0.02188 (83.137)***	0.02188 (83.137)***	0.02188 (83.137)***	0.02188 (83.137)***	0.03385 (15.580)***	0.03385 (15.580)***	0.03385 (15.580)***	0.03385 (15.580)***	0.03385 (15.580)***	0.03385 (15.580)***	0.03385 (15.580)***	0.03385 (15.580)***
$ASC_{i,t-1}$	0.08150 (47.139)***	0.08150 (47.139)***	0.08150 (47.139)***	0.08150 (47.139)***	0.17361 (8.489)***	0.17361 (8.489)***	0.17361 (8.489)***	0.17361 (8.489)***	0.17361 (8.489)***	0.17361 (8.489)***	0.17361 (8.489)***	0.17361 (8.489)***
$OPC_{i,t-1}$	0.00396 (12.161)***	0.00396 (12.161)***	0.00396 (12.161)***	0.00396 (12.161)***	-0.02054 (3.042)***	-0.02054 (3.042)***	-0.02054 (3.042)***	-0.02054 (3.042)***	-0.03523 (3.550)***	-0.03607 (3.642)***	-0.03607 (3.642)***	-0.03607 (3.642)***
$IC_{i,t-1}$	0.02294 (12.870)***	0.02124 (12.081)***	0.02142 (12.161)***	0.02155 (12.155)**	-0.02092 (3.084)***	-0.02054 (3.042)***	-0.02054 (3.042)***	-0.02054 (3.042)***	-0.03523 (3.550)***	-0.03607 (3.642)***	-0.03607 (3.642)***	-0.03607 (3.642)***
$Intercept$	0.02294 (12.870)***	0.02124 (12.081)***	0.02142 (12.161)***	0.02155 (12.155)**	-0.02092 (3.084)***	-0.02054 (3.042)***	-0.02054 (3.042)***	-0.02054 (3.042)***	-0.03523 (3.550)***	-0.03607 (3.642)***	-0.03607 (3.642)***	-0.03607 (3.642)***
R^2	0.18 Yes	0.18 Yes	0.18 Yes	0.50 Yes	0.02 Yes	0.02 Yes	0.02 Yes	0.03 Yes	0.05 Yes	0.05 Yes	0.05 Yes	0.03 Yes
Currency and Time FE	672,569	681,229	680,255	107,410	267,513	268,489	268,487	68,732	84,022	84,155	84,156	27,259
N												

Table 13: This table presents the results of the following regression: $SLOPE_{i,t}^A = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \epsilon_{i,t}$. We restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse. Panel A, B, C present the results of the estimations for time intervals set to 1 minute, 15 minutes and 60 minutes, respectively. Respective columns in each panel are for the regressions where the components of the spread ASC^k , OPC^k are obtained by [Glosten and Harris \(1988\)](#) ($k = GH$), two-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS2 - way$), [Madhavan et al. \(1997\)](#) ($k = MRR$), and three-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS3 - way$). Inventory costs (IC) are obtained only by the three-way decomposition of [Huang and Stoll \(1997\)](#). We require at least 30 trades within the time interval t , to estimate the adverse selection component in that interval. Absolute value of the t -statistics in parenthesis. Hypothesis 2 asserts a relationship $\gamma_6 > 0$. ***, **, * indicates statistical significance at 1% , 5% and 10%, respectively.

	Panel A: 1 min				Panel B: 15 min				Panel C: 60 min			
	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way
$\sigma_{i,t-1}^R$	-2.82754 (3.069)***	-3.07446 (3.226)***	-3.06361 (3.211)***	-1.82046 (4.650)***	-1.43004 (6.501)***	-1.44275 (6.581)***	-1.44278 (6.581)***	-1.45626 (4.493)***	-0.71019 (2.356)**	-0.71467 (2.373)**	-0.71467 (2.373)**	-0.47521 (1.836)*
$SPREAD_{i,t-1}$	0.00150 (18.672)***	0.00151 (18.886)***	0.00151 (18.870)***	0.00141 (16.583)***	0.00076 (5.649)***	0.00071 (5.465)***	0.00071 (5.465)***	0.00012 (0.854)	0.00106 (2.141)**	0.00105 (2.137)**	0.00105 (2.137)**	0.00118 (1.998)**
$TV_{i,t-1}$	-0.00596 (3.931)***	-0.00568 (3.808)***	-0.00572 (3.825)***	-0.00597 (9.603)***	0.00018 (0.274)	0.00025 (0.372)	0.00025 (0.371)	-0.00047 (0.489)	-0.00125 (0.766)	-0.00121 (0.745)	-0.00121 (0.745)	-0.00166 (1.140)
$BSI_{i,t-1}$	0.00019 (3.625)***	0.00018 (3.576)***	0.00018 (3.588)***	0.00012 (5.960)***	0.00003 (5.243)***	0.00003 (5.220)***	0.00003 (5.222)***	0.00003 (3.763)***	0.00001 (2.308)**	0.00001 (2.317)**	0.00001 (2.317)**	0.00001 (2.729)***
$MT_{i,t-1}$	-0.02879 (4.468)***	-0.02743 (4.331)***	-0.02757 (4.343)***	-0.02746 (10.510)***	-0.00797 (2.336)**	-0.00771 (2.277)**	-0.00772 (2.279)**	-0.01022 (2.167)**	-0.00395 (0.407)	-0.00383 (0.396)	-0.00383 (0.396)	0.00715 (0.853)
$ASC_{i,t-1}$	0.15639 (28.007)***				0.20906 (25.708)***				0.30740 (11.352)***			
$OPC_{i,t-1}$	0.01556 (17.719)***				0.03186 (38.712)***				0.03528 (15.891)***			
$ASC_{i,t-1}$		0.16132 (28.720)***				0.21257 (26.133)***				0.31261 (11.541)***		
$OPC_{i,t-1}$		0.01559 (17.741)***				0.03184 (38.584)***				0.03510 (15.760)***		
$ASC_{i,t-1}$			0.16138 (28.704)***				0.21259 (26.135)***					
$OPC_{i,t-1}$			(17.725)***				(38.584)***					
$ASC_{i,t-1}$				0.05512 (21.917)***				0.11548 (9.990)***				0.14751 (5.761)***
$OPC_{i,t-1}$				0.01769 (35.291)***				0.03350 (21.624)***				0.03173 (12.299)***
$ASC_{i,t-1}$				0.08835 (26.819)***				0.18715 (12.837)***				0.25066 (8.497)***
$IC_{i,t-1}$				0.01991 (5.691)***				-0.02344 (3.603)***				-0.02389 (2.052)**
$Intercept$	0.00423 (0.503)	0.00278 (0.336)	0.00290 (0.350)		-0.02880 (6.489)***	-0.02938 (6.650)***	-0.02938 (6.651)***		-0.03533 (2.773)***	-0.03602 (2.833)***	-0.03602 (2.833)***	
R^2	0.01 Yes	0.01 Yes	0.01 Yes	0.21 Yes	0.06 Yes	0.06 Yes	0.06 Yes	0.08 Yes	0.04 Yes	0.04 Yes	0.04 Yes	0.06 Yes
Currency and Time FE	674,312	682,991	682,020	107,673	267,925	268,902	268,901	68,839	84,318	84,451	84,452	27,357
N												

Table 14: This table presents the results of the following regression: $DEPTH_{i,t}^B = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \epsilon_{i,t}$. We restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse. Panel A, B, C present the results of the estimations for time intervals set to 1 minute, 15 minutes and 60 minutes, respectively. Respective columns in each panel are for the regressions where the components of the spread ASC^k , OPC^k are obtained by [Glosten and Harris \(1988\)](#) ($k = GH$), two-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS2 - way$), [Madhavan et al. \(1997\)](#) ($k = MRR$), and three-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS3 - way$). Inventory costs (IC) are obtained only by the three-way decomposition of [Huang and Stoll \(1997\)](#). We require at least 30 trades within the time interval t , to estimate the adverse selection component in that interval. Absolute value of the t -statistics in parenthesis. Hypothesis 2 asserts a relationship $\gamma_6 < 0$. ***, **, * indicates statistical significance at 1% , 5% and 10%, respectively.

	Panel A: 1 min				Panel B: 15 min				Panel C: 60 min			
	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way
$\sigma_{i,t-1}^R$	52.43454 (2.105)*	60.34179 (2.357)*	60.14232 (2.346)*	26.04875 (0.243)	12.10289 (2.898)**	11.62395 (2.784)**	11.64354 (2.788)**	21.28383 (2.673)**	3.61126 (1.319)	3.68521 (1.349)	3.68426 (1.348)	2.96110 (0.743)
$SPREAD_{i,t-1}$	0.00011 (0.053)	0.00012 (0.055)	0.00012 (0.055)	0.00123 (0.053)	0.00008 (0.033)	0.00005 (0.021)	0.00005 (0.021)	0.00007 (0.021)	-0.00017 (0.038)	-0.00018 (0.039)	-0.00018 (0.039)	-0.00063 (0.070)
$TV_{i,t-1}$	0.14494 (3.534)**	0.13739 (3.426)**	0.13759 (3.423)**	0.24887 (1.460)	0.07877 (6.159)**	0.07994 (6.262)**	0.07978 (6.251)**	0.07605 (3.199)**	0.03181 (2.143)*	0.03078 (2.080)*	0.03079 (2.081)*	0.11311 (5.044)**
$BSI_{i,t-1}$	-0.00390 (2.771)**	-0.00382 (2.775)**	-0.00382 (2.766)**	-0.00218 (0.396)	-0.00119 (11.026)**	-0.00118 (11.015)**	-0.00118 (11.003)**	-0.00152 (7.591)**	-0.00040 (8.843)**	-0.00040 (8.868)**	-0.00040 (8.867)**	-0.00050 (6.461)**
$MT_{i,t-1}$	0.48902 (2.808)**	0.48116 (2.827)**	0.48341 (2.834)**	1.03464 (1.445)	0.87031 (13.438)**	0.86521 (13.414)**	0.86365 (13.387)**	0.94730 (8.175)**	1.04473 (11.863)**	1.03070 (11.756)**	1.03038 (11.754)**	1.05562 (8.179)**
$ASC_{i,t-1}$	0.00801 (0.053)				0.05893 (0.382)				0.16218 (0.660)			
$OPC_{i,t-1}$	-0.00710 (0.299)				0.01496 (0.957)				0.02914 (1.446)			
$ASC_{i,t-1}$	0.00349 (0.023)				0.06462 (0.417)				0.16029 (0.652)			
$OPC_{i,t-1}$	-0.00796 (0.337)				0.01493 (0.950)				0.02919 (1.445)			
$ASC_{i,t-1}$		0.00375 (0.025)				0.06423 (0.415)				0.16027 (0.652)		
$OPC_{i,t-1}$		-0.00794 (0.336)				0.01488 (0.947)				0.02918 (1.444)		
$ASC_{i,t-1}$				-0.02323 (0.034)				0.05023 (0.177)				0.33132 (0.841)
$OPC_{i,t-1}$				0.00729 (0.053)				0.02248 (0.591)				0.00456 (0.115)
$ASC_{i,t-1}$				-0.10774 (0.119)				0.03252 (0.091)				0.45413 (1.000)
$IC_{i,t-1}$				-1.53546 (1.601)				-0.65441 (4.094)**				-0.98827 (5.516)**
$Intercept$	-0.81855 (3.599)**	-0.79133 (3.556)**	-0.79282 (3.557)**		-0.63169 (7.497)**	-0.63700 (7.570)**	-0.63576 (7.556)**		-0.39184 (3.388)**	-0.38216 (3.314)**	-0.38218 (3.314)**	
R^2	0.01 Yes	0.01 Yes	0.01 Yes	0.00 Yes	0.20 Yes	0.20 Yes	0.20 Yes	0.24 Yes	0.40 Yes	0.40 Yes	0.40 Yes	0.48 Yes
Currency and Time FE	674,322	683,001	682,030	107,673	267,925	268,902	268,901	68,839	84,318	84,451	84,452	27,357
N												

Table 15: This table presents the results of the following regression: $DEPTH_{i,t}^A = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \epsilon_{i,t}$. We restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse. Panel A, B, C present the results of the estimations for time intervals set to 1 minute, 15 minutes and 60 minutes, respectively. Respective columns in each panel are for the regressions where the components of the spread ASC^k , OPC^k are obtained by [Glosten and Harris \(1988\)](#) ($k = GH$), two-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS2 - way$), [Madhavan et al. \(1997\)](#) ($k = MRR$), and three-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS3 - way$). Inventory costs (IC) are obtained only by the three-way decomposition of [Huang and Stoll \(1997\)](#). We require at least 30 trades within the time interval t , to estimate the adverse selection component in that interval. Absolute value of the t -statistics are in parenthesis. Hypothesis 2 asserts a relationship $\gamma_6 < 0$. ***, * indicates statistical significance at 1% , 5% and 10%, respectively.

	Panel A: 1 min				Panel B: 15 min				Panel C: 60 min			
	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way
$\sigma_{i,t-1}^R$	33.78676 (3.783)**	37.67565 (4.127)**	37.97311 (4.156)**	38.46850 (1.765)	15.84417 (3.013)**	16.36739 (3.107)**	16.37609 (3.109)**	10.82265 (1.947)	6.05765 (1.539)	5.96387 (1.517)	5.96320 (1.517)	-6.07019 (0.576)
$SPREAD_{i,t-1}$	0.00007 (0.095)	0.00008 (0.105)	0.00008 (0.106)	0.00074 (0.157)	0.00004 (0.012)	0.00003 (0.008)	0.00003 (0.008)	0.00009 (0.035)	-0.00024 (0.038)	-0.00025 (0.039)	-0.00025 (0.039)	-0.00052 (0.021)
$TV_{i,t-1}$	0.16771 (11.408)**	0.16022 (11.207)**	0.16094 (11.235)**	0.20382 (5.885)**	0.07716 (4.791)**	0.07583 (4.710)**	0.07573 (4.704)**	0.10313 (6.214)**	0.03552 (1.664)	0.03760 (1.766)	0.03761 (1.766)	0.12660 (2.133)*
$BSI_{i,t-1}$	-0.00365 (7.231)**	-0.00351 (7.151)**	-0.00349 (7.083)**	-0.00337 (3.016)**	-0.00102 (7.536)**	-0.00104 (7.639)**	-0.00104 (7.636)**	-0.00104 (7.417)**	-0.00027 (4.188)**	-0.00028 (4.235)**	-0.00028 (4.235)**	-0.00038 (1.880)
$MT_{i,t-1}$	0.44200 (7.079)**	0.43039 (7.093)**	0.42219 (6.943)**	0.27096 (1.863)	0.79404 (9.736)**	0.80936 (9.948)**	0.80886 (9.940)**	0.74269 (9.182)**	0.73379 (5.793)**	0.74538 (5.908)**	0.74516 (5.907)**	0.53323 (1.561)
$ASC_{i,t-1}$	0.02253 (0.416)				0.05437 (0.280)				0.12703 (0.359)			
$OPC_{i,t-1}$	-0.00848 (0.996)				0.00565 (0.287)				0.00749 (0.258)			
$ASC_{i,t-1}$		0.01873 (0.348)				0.05573 (0.285)				0.13700 (0.387)		
$OPC_{i,t-1}$		-0.00854 (1.014)				0.00620 (0.313)				0.00712 (0.245)		
$ASC_{i,t-1}$			0.01919 (0.356)				0.05530 (0.283)				0.13699 (0.387)	
$OPC_{i,t-1}$			-0.00884 (1.049)				0.00619 (0.312)				0.00711 (0.245)	
$ASC_{i,t-1}$				0.02538 (0.181)				0.10116 (0.510)				0.13416 (0.129)
$OPC_{i,t-1}$				-0.01309 (0.469)				0.00042 (0.016)				0.00614 (0.058)
$IC_{i,t-1}$				0.00813 (0.044)				0.13872 (0.555)				0.27325 (0.227)
$Intercept$	-0.87112 (10.683)**	-0.84234 (10.617)**	-0.84338 (10.616)**	-0.98281 (5.046)**	-0.61951 (5.839)**	-0.61700 (5.813)**	-0.61629 (5.807)**	-0.76177 (6.827)**	-0.38161 (2.294)*	-0.39835 (2.400)*	-0.39837 (2.400)*	-0.93147 (1.964)*
R^2	0.06	0.06	0.06	0.07	0.12	0.12	0.12	0.35	0.22	0.22	0.22	0.10
Currency and Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	674,322	683,001	682,030	107,673	267,925	268,902	268,901	68,839	84,318	84,451	84,452	27,357

Table 16: This table presents the results of the following regression: $MT_{i,t} = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BS I_{i,t-1} + \gamma_5 MT_{i,t-1} + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \epsilon_{i,t}$. We restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse. Panel A, B, C present the results of the estimation for time intervals set to 1 minute, 15 minutes and 60 minutes, respectively. Respective columns in each panel are for the regressions where the components of the spread ASC^k , OPC^k are obtained by [Glosten and Harris \(1988\)](#) ($k = GH$), two-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS2 - way$), [Madhavan et al. \(1997\)](#) ($k = MRR$), and three-way decomposition of [Huang and Stoll \(1997\)](#) ($k = HS3 - way$). Inventory costs (IC) are obtained only by the three-way decomposition of [Huang and Stoll \(1997\)](#). We require at least 30 trades within the time interval t , to estimate the adverse selection component in that interval. Absolute value of the t -statistics are in parenthesis. Hypothesis 3 asserts a relationship $\gamma_6 > 0$. ***, **, * indicates statistical significance at 1% , 5% and 10%, respectively.

	Panel A: 1 min				Panel B: 15 min				Panel C: 60 min			
	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way	GH	HS 2-way	MRR	HS 3-way
$\sigma_{i,t-1}^R$	-9.45390 (41.816)***	-10.62251 (44.690)***	-10.51745 (44.266)***	-10.67045 (17.482)***	-4.12059 (29.359)***	-4.11200 (29.324)***	-4.12092 (29.386)***	-4.86010 (16.375)***	-1.54267 (12.905)***	-1.54980 (12.954)***	-1.54951 (12.951)***	-1.64179 (7.836)***
$SPREAD_{i,t-1}$	-0.00012 (4.081)***	-0.00012 (4.045)***	-0.00012 (4.039)***	-0.00027 (2.266)***	-0.00025 (3.008)***	-0.00025 (3.005)***	-0.00025 (3.006)***	-0.00015 (1.132)	-0.00027 (1.378)	-0.00027 (1.369)	-0.00027 (1.369)	0.00019 (0.394)
$TV_{i,t-1}$	-0.01174 (28.654)***	-0.01205 (29.612)***	-0.01187 (29.136)***	-0.01419 (13.259)***	-0.02106 (47.673)***	-0.02111 (47.888)***	-0.02108 (47.827)***	-0.02056 (22.632)***	-0.02088 (32.033)***	-0.02090 (32.083)***	-0.02090 (32.088)***	-0.01968 (16.608)***
$BSI_{i,t-1}$	-0.00031 (24.497)***	-0.00029 (22.754)***	-0.00029 (23.068)***	-0.00024 (7.615)***	0.00000 (0.501)	0.00000 (0.602)	0.00000 (0.616)	0.00001 (0.908)	0.00001 (2.628)***	0.00001 (2.689)***	0.00001 (2.680)***	0.00001 (1.941)*
$MT_{i,t-1}$	0.22503 (124.219)***	0.22834 (127.007)***	0.22639 (125.815)***	0.20444 (44.341)***	0.18082 (79.417)***	0.18243 (80.486)***	0.18251 (80.502)***	0.18171 (40.440)***	0.20338 (52.543)***	0.20429 (52.871)***	0.20439 (52.901)***	0.20935 (30.718)***
$ASC_{i,t-1}$	-0.03920 (28.606)***				-0.12782 (23.995)***				-0.22914 (21.269)***			
$OPC_{i,t-1}$	-0.00467 (22.434)***				-0.00401 (7.623)***				0.00164 (1.863)*			
$ASC_{i,t-1}$	-0.04090 (29.511)***				-0.13192 (24.712)***					-0.23250 (21.536)***		
$OPC_{i,t-1}$	-0.00455 (21.675)***				-0.00384 (7.260)***					0.00185 (2.087)**		
$ASC_{i,t-1}$	-0.04039 (29.161)***				-0.13204 (24.733)***					-0.23250 (21.535)***		
$OPC_{i,t-1}$	-0.00455 (21.699)***				-0.00384 (7.266)***					0.00185 (2.091)**		
$ASC_{i,t-1}$	-0.02915 (7.396)***							-0.08390 (7.708)***				-0.13607 (6.516)***
$OPC_{i,t-1}$	-0.00522 (6.964)***							-0.00543 (3.816)***				0.00063 (0.303)
$ASC_{i,t-1}$	-0.05303 (10.446)***							(9.757)***				-0.25407 (10.611)***
$IC_{i,t-1}$	0.37491 (159.292)***	0.37757 (161.522)***	0.37684 (161.188)***	0.40211 (64.801)***	0.34710 (119.090)***	0.34744 (119.434)***	0.34732 (119.388)***	0.35024 (57.260)***	0.31594 (62.221)***	0.31629 (62.314)***	0.31629 (62.314)***	0.31266 (33.001)***
R^2	0.09 Yes	0.09 Yes	0.09 Yes	0.09 Yes	0.12 Yes	0.12 Yes	0.12 Yes	0.12 Yes	0.14 Yes	0.14 Yes	0.14 Yes	0.13 Yes
Currency and Time FE	488,796	494,244	493,641	74,528	254,801	255,657	255,660	65,729	83,808	83,941	83,942	27,231
N												

Table 17: $R_{i,t} = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} + \gamma_6 ASC_{i,t-1} + \gamma_7 OPC_{i,t-1} + \gamma_8 IC_{i,t-1} + \gamma_9 DEPTH_{i,t-1} + \gamma_{10} SLOPE_{i,t-1} + \epsilon_{i,t}$. We restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse. The components of the spread are estimated using the [Glosten and Harris \(1988\)](#) model. Panel A and B present the results of the estimation for time intervals set to 60 minutes and 1 minutes, respectively. We require at least 30 trades within the time interval t , to estimate the adverse selection component in that interval. Absolute value of the t -statistics are in parenthesis. Hypothesis 4 asserts a relationship $\gamma_6 \neq 0$. ***, * indicates statistical significance at 1% , 5% and 10%, respectively.

	Panel A: 1min						Panel B: 60min					
	1	2	3	4	5	6	1	2	3	4	5	6
$ASC_{i,t-1}$	-0.00003 (1.255)	-0.00006 (2.463)*	-0.00006 (2.513)*	-0.00006 (2.420)*	-0.00006 (2.422)*	-0.00006 (2.617)**	-0.00298 (2.189)*	-0.00270 (1.950)	-0.00283 (2.031)*	-0.00326 (2.179)*	-0.00321 (2.142)*	-0.00288 (2.025)*
$OPC_{i,t-1}$	0.00002 (5.297)**	0.00000 (0.342)	0.00000 (0.386)	0.00000 (0.521)	0.00000 (0.516)	0.00000 (0.319)	0.00009 (0.857)	-0.00016 (1.429)	-0.00016 (1.431)	-0.00020 (1.601)	-0.00018 (1.426)	-0.00015 (1.273)
$\sigma_{i,t-1}^R$		0.08694 (21.868)**	0.08623 (21.588)**	0.08633 (21.328)**	0.08639 (21.344)**	0.08659 (21.400)**		0.22352 (10.800)**	0.21638 (10.402)**	0.22559 (10.730)**	0.22541 (10.722)**	0.22436 (10.699)**
$ILLIQ_{i,t-1}$		0.04362 (5.727)**	0.04444 (5.745)**	0.04287 (5.406)**	0.04284 (5.402)**	0.04281 (5.399)**		-0.51977 (7.520)**	-0.48542 (6.952)**	-0.50568 (7.158)**	-0.50475 (7.144)**	-0.50434 (7.139)**
$BSI_{i,t-1}$		-0.00000 (2.152)*	-0.00000 (2.252)*	-0.00000 (2.308)*	-0.00000 (2.309)*	-0.00000 (2.361)*		0.00000 (0.574)	0.00000 (0.602)	0.00000 (0.503)	0.00000 (0.468)	0.00000 (0.518)
$TV_{i,t-1}$		0.00001 (2.452)*	0.00002 (2.487)*	0.00002 (2.397)*	0.00002 (2.399)*	0.00002 (2.470)*		0.00006 (0.764)	0.00007 (0.775)	0.00005 (0.566)	0.00005 (0.594)	0.00005 (0.613)
$MT_{i,t-1}$		0.00006 (2.323)*	0.00006 (2.188)*	0.00006 (2.072)*	0.00006 (2.085)*	0.00006 (2.135)*		0.00064 (1.272)	0.00056 (1.110)	0.00053 (1.044)	0.00053 (1.042)	0.00052 (1.018)
$DEPTH_{i,t-1}$			-0.00000 (0.478)	-0.00000 (0.485)	-0.00000 (0.485)				0.00002 (1.905)	0.00002 (1.760)	0.00002 (1.759)	
$SLOPE_{i,t-1}$				-0.00001 (0.260)	-0.00002 (0.279)					0.00114 (0.789)	0.00127 (0.876)	
$SPREAD_{i,t-1}$					0.00000 (0.354)	0.00000 (0.327)					-0.00006 (2.160)*	-0.00006 (2.138)*
$SLOPE_{i,t-1}^B$						0.00002 (0.956)						0.00018 (0.770)
$SLOPE_{i,t-1}^A$						0.00000 (0.245)						0.00011 (0.605)
$DEPTH_{i,t-1}^B$						0.00000 (0.097)						0.00003 (1.519)
$DEPTH_{i,t-1}^A$						-0.00000 (1.698)						0.00001 (0.983)
$Intercept$	0.00001 (0.673)	-0.00022 (6.417)**	-0.00022 (6.358)**	-0.00022 (6.162)**	-0.00022 (6.168)**	-0.00022 (6.215)**	0.00047 (1.687)	-0.00048 (0.731)	-0.00048 (0.730)	-0.00029 (0.427)	-0.00025 (0.377)	-0.00030 (0.453)
R^2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Currency and Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	700,743	700,702	693,031	674,343	674,285	674,247	87,016	87,004	86,109	84,308	84,306	84,306

Table 18: $R_{i,t} = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} + \gamma_6 ASC_{i,t-1} + \gamma_7 OPC_{i,t-1} + \gamma_8 IC_{i,t-1} + \gamma_9 DEPTH_{i,t-1} + \gamma_{10} SLOPE_{i,t-1} + \epsilon_{i,t}$. We restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse. The components of the spread are estimated using the [Madhavan et al. \(1997\)](#) model. Panel A and B present the results of the estimation for time intervals set to 60 minutes and 1 minutes, respectively. We require at least 30 trades within the time interval t , to estimate the adverse selection component in that interval. Absolute value of the t -statistics are in parenthesis. Hypothesis 4 asserts a relationship $\gamma_6 \neq 0$. ***, **, * indicates statistical significance at 1% , 5% and 10%, respectively.

	Panel A: 1min						Panel B: 60min					
	1	2	3	4	5	6	1	2	3	4	5	6
$ASC_{i,t-1}$	-0.00001 (0.550)	-0.00005 (2.153)*	-0.00005 (2.183)*	-0.00005 (2.148)*	-0.00005 (2.149)*	-0.00005 (2.287)*	-0.00305 (2.244)*	-0.00279 (2.015)*	-0.00294 (2.112)*	-0.00340 (2.266)*	-0.00334 (2.228)*	-0.00299 (2.103)*
$OPC_{i,t-1}$	0.00002 (4.463)**	-0.00000 (0.824)	-0.00000 (0.779)	-0.00000 (0.815)	-0.00000 (0.823)	-0.00000 (0.983)	0.00010 (0.881)	-0.00016 (1.390)	-0.00016 (1.386)	-0.00020 (1.573)	-0.00018 (1.398)	-0.00014 (1.229)
$\sigma_{i,t-1}^R$	0.11542 (28.514)**	0.11496 (28.281)**	0.11288 (26.574)**	0.11296 (26.642)**	0.11296 (26.591)**	0.11313 (26.642)**	0.22354 (10.807)**	0.22354 (10.807)**	0.21660 (10.419)**	0.22600 (10.757)**	0.22582 (10.749)**	0.22472 (10.723)**
$ILLIQ_{i,t-1}$	0.01542 (2.720)**	0.01531 (2.678)**	0.02635 (3.324)**	0.02626 (3.313)**	0.02628 (3.315)**	0.02626 (3.313)**	-0.51946 (7.524)**	-0.51946 (7.524)**	-0.48554 (6.962)**	-0.50679 (7.182)**	-0.50588 (7.169)**	-0.50549 (7.163)**
$BSI_{i,t-1}$	-0.00000 (3.076)**	-0.00000 (3.176)**	-0.00000 (3.361)**	-0.00000 (3.361)**	-0.00000 (3.364)**	-0.00000 (3.398)**	0.00000 (3.398)**	0.00000 (3.398)**	0.00000 (3.398)**	0.00000 (3.398)**	0.00000 (3.398)**	0.00000 (3.398)**
$TV_{i,t-1}$	0.00001 (1.039)	0.00001 (1.077)	0.00001 (1.139)	0.00001 (1.139)	0.00001 (1.143)	0.00001 (1.182)	0.00007 (0.786)	0.00007 (0.786)	0.00007 (0.786)	0.00005 (0.573)	0.00005 (0.600)	0.00005 (0.621)
$MT_{i,t-1}$	0.00009 (3.341)**	0.00009 (3.187)**	0.00008 (3.011)**	0.00008 (3.011)**	0.00008 (3.028)**	0.00008 (3.058)**	0.00067 (1.331)	0.00067 (1.331)	0.00058 (1.160)	0.00055 (1.088)	0.00055 (1.087)	0.00054 (1.062)
$DEPTH_{i,t-1}$	-0.00000 (0.348)	-0.00000 (0.354)	-0.00000 (0.354)	-0.00000 (0.354)	-0.00000 (0.354)	-0.00000 (0.354)	0.00000 (0.354)	0.00000 (0.354)	0.00000 (0.354)	0.00002 (1.762)	0.00002 (1.762)	0.00002 (1.762)
$SLOPE_{i,t-1}$	0.00001 (0.093)	0.00001 (0.093)	0.00001 (0.093)	0.00001 (0.093)	0.00001 (0.093)	0.00001 (0.093)	0.00001 (0.093)	0.00001 (0.093)	0.00001 (0.093)	0.00120 (0.828)	0.00133 (0.915)	0.00133 (0.915)
$SPREAD_{i,t-1}$	0.00000 (0.435)	0.00000 (0.435)	0.00000 (0.435)	0.00000 (0.435)	0.00000 (0.435)	0.00000 (0.435)	0.00000 (0.435)	0.00000 (0.435)	0.00000 (0.435)	0.00000 (0.435)	0.00000 (0.435)	0.00000 (0.435)
$SLOPE_{i,t-1}^B$	0.00003 (1.174)	0.00003 (1.174)	0.00003 (1.174)	0.00003 (1.174)	0.00003 (1.174)	0.00003 (1.174)	0.00003 (1.174)	0.00003 (1.174)	0.00003 (1.174)	0.00018 (0.771)	0.00018 (0.771)	0.00018 (0.771)
$SLOPE_{i,t-1}^A$	0.00011 (0.605)	0.00011 (0.605)	0.00011 (0.605)	0.00011 (0.605)	0.00011 (0.605)	0.00011 (0.605)	0.00011 (0.605)	0.00011 (0.605)	0.00011 (0.605)	0.00011 (0.605)	0.00011 (0.605)	0.00011 (0.605)
$DEPTH_{i,t-1}^B$	0.00003 (1.523)	0.00003 (1.523)	0.00003 (1.523)	0.00003 (1.523)	0.00003 (1.523)	0.00003 (1.523)	0.00003 (1.523)	0.00003 (1.523)	0.00003 (1.523)	0.00003 (1.523)	0.00003 (1.523)	0.00003 (1.523)
$DEPTH_{i,t-1}^A$	0.00001 (0.982)	0.00001 (0.982)	0.00001 (0.982)	0.00001 (0.982)	0.00001 (0.982)	0.00001 (0.982)	0.00001 (0.982)	0.00001 (0.982)	0.00001 (0.982)	0.00001 (0.982)	0.00001 (0.982)	0.00001 (0.982)
$Intercept$	0.00002 (1.114)	-0.00020 (5.796)**	-0.00019 (5.724)**	-0.00020 (5.783)**	-0.00020 (5.785)**	-0.00020 (5.811)**	0.00048 (1.701)	-0.00049 (0.749)	-0.00049 (0.734)	-0.00028 (0.423)	-0.00025 (0.373)	-0.00030 (0.454)
R^2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Currency and Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	708,748	708,709	700,954	682,053	681,995	681,954	87,153	87,141	86,245	84,442	84,440	84,440

Table 19: $R_{i,t} = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} + \gamma_6 ASC_{i,t-1} + \gamma_7 OPC_{i,t-1} + \gamma_8 IC_{i,t-1} + \gamma_9 DEPTH_{i,t-1} + \gamma_{10} SLOPE_{i,t-1} + \epsilon_{i,t}$. We restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse. The components of the spread are estimated using the two-way decomposition of [Huang and Stoll \(1997\)](#) model. Panel A and B present the results of the estimation for time intervals set to 60 minutes and 1 minutes, respectively. We require at least 30 trades within the time interval t , to estimate the adverse selection component in that interval. Absolute value of the t -statistics are in parenthesis. Hypothesis 4 asserts a relationship $\gamma_6 \neq 0$. ***, **, * indicates statistical significance at 1%, 5% and 10%, respectively.

	Panel A: 1min						Panel B: 60min					
	1	2	3	4	5	6	1	2	3	4	5	6
$ASC_{i,t-1}$	-0.00001 (0.564)	-0.00005 (2.177)*	-0.00005 (2.210)*	-0.00005 (2.169)*	-0.00005 (2.171)*	-0.00005 (2.311)*	-0.00305 (2.244)*	-0.00279 (2.015)*	-0.00294 (2.112)*	-0.00340 (2.266)*	-0.00334 (2.228)*	-0.00299 (2.103)*
$OPC_{i,t-1}$	0.00002 (4.447)**	-0.00000 (0.845)	-0.00000 (0.800)	-0.00000 (0.824)	-0.00000 (0.832)	-0.00000 (0.996)	0.00010 (0.881)	-0.00016 (1.390)	-0.00016 (1.386)	-0.00020 (1.573)	-0.00018 (1.398)	-0.00014 (1.229)
$\sigma_{i,t-1}^R$	0.11563 (28.613)**	0.11512 (28.369)**	0.11512 (28.369)**	0.11319 (26.722)**	0.11326 (26.739)**	0.11344 (26.791)**	0.22354 (10.807)**	0.22354 (10.807)**	0.21660 (10.419)**	0.22600 (10.757)**	0.22582 (10.749)**	0.22472 (10.722)**
$ILLIQ_{i,t-1}$	0.01495 (2.660)**	0.01498 (2.641)**	0.01498 (2.641)**	0.02545 (3.258)**	0.02538 (3.249)**	0.02536 (3.247)**	-0.51946 (7.524)**	-0.51946 (7.524)**	-0.48554 (6.961)**	-0.50679 (7.182)**	-0.50588 (7.169)**	-0.50549 (7.163)**
$BSI_{i,t-1}$	-0.00000 (3.124)**	-0.00000 (3.124)**	-0.00000 (3.213)**	-0.00000 (3.414)**	-0.00000 (3.417)**	-0.00000 (3.451)**	0.00000 (3.451)**	0.00000 (3.451)**	0.00000 (3.451)**	0.00000 (3.451)**	0.00000 (3.451)**	0.00000 (3.451)**
$TV_{i,t-1}$	0.00001 (1.034)	0.00001 (1.034)	0.00001 (1.066)	0.00001 (1.130)	0.00001 (1.134)	0.00001 (1.174)	0.00007 (0.786)	0.00007 (0.786)	0.00007 (0.786)	0.00005 (0.573)	0.00005 (0.600)	0.00005 (0.621)
$MT_{i,t-1}$	0.00009 (3.317)**	0.00009 (3.317)**	0.00008 (3.162)**	0.00008 (3.002)**	0.00008 (3.018)**	0.00008 (3.049)**	0.00067 (1.330)	0.00067 (1.330)	0.00058 (1.160)	0.00055 (1.088)	0.00055 (1.087)	0.00054 (1.062)
$DEPTH_{i,t-1}$	-0.00000 (0.358)	-0.00000 (0.358)	-0.00000 (0.364)	-0.00000 (0.364)	-0.00000 (0.364)	-0.00000 (0.414)	0.00002 (1.907)	0.00002 (1.907)	0.00002 (1.907)	0.00002 (1.762)	0.00002 (1.762)	0.00002 (1.762)
$SLOPE_{i,t-1}$	0.00001 (0.074)	0.00001 (0.074)	0.00001 (0.074)	0.00001 (0.096)	0.00000 (0.074)	0.00000 (0.431)	0.00133 (0.915)	0.00133 (0.915)	0.00120 (0.828)	0.00120 (0.828)	0.00133 (0.915)	-0.00006 (2.136)*
$SPREAD_{i,t-1}$	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)
$SLOPE_{i,t-1}^B$	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)
$SLOPE_{i,t-1}^A$	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)
$DEPTH_{i,t-1}^B$	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)
$DEPTH_{i,t-1}^A$	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)	0.00000 (0.431)
$Intercept$	0.00002 (1.131)	-0.00019 (5.770)**	-0.00019 (5.695)**	-0.00020 (5.751)**	-0.00020 (5.753)**	-0.00020 (5.779)**	0.00048 (1.701)	-0.00049 (0.749)	-0.00049 (0.734)	-0.00028 (0.423)	-0.00025 (0.373)	-0.00030 (0.982)
R^2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Currency and Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	709,774	709,735	701,960	683,026	682,966	682,925	87,152	87,140	86,244	84,441	84,439	84,439

Table 20: $R_{i,t} = \alpha_i + \theta_t + \gamma_1 \sigma_{i,t-1}^R + \gamma_2 SPREAD_{i,t-1} + \gamma_3 TV_{i,t-1} + \gamma_4 BSI_{i,t-1} + \gamma_5 MT_{i,t-1} + \gamma_6 ASC_{i,t-1}^k + \gamma_7 OPC_{i,t-1}^k + \gamma_8 IC_{i,t-1}^k + \gamma_9 DEPTH_{i,t-1} + \gamma_{10} SLOPE_{i,t-1} + \epsilon_{i,t}$. We restrict our sample to the period between August 2017 and June 2018 in order to provide a substantial sample of cryptocurrencies upon which the market could trade, therefore providing a strong sample to analyse. The components of the spread are estimated using the three-way decomposition of [Huang and Stoll \(1997\)](#) model. Panel A and B present the results of the estimation for time intervals set to 60 minutes and 1 minutes, respectively. We require at least 30 trades within the time interval t , to estimate the adverse selection component in that interval. Absolute value of the t -statistics are in parenthesis. Hypothesis 4 asserts a relationship $\gamma_6 \neq 0$. ***, **, * indicates statistical significance at 1% , 5% and 10%, respectively.

	Panel A: 1 min								
	1	2	3	4	5	6	7	8	9
$ASC_{i,t-1}$	0.00002 (0.411)			0.00006 (1.052)	0.00004 (0.739)	0.00004 (0.624)	0.00005 (0.727)	0.00005 (0.721)	0.00004 (0.711)
$IC_{i,t-1}$		0.00003 (0.426)		0.00008 (1.040)	0.00007 (0.891)	0.00007 (0.920)	0.00008 (0.939)	0.00008 (0.938)	0.00008 (0.930)
$OPC_{i,t-1}$			-0.00002 (2.049)*	-0.00003 (2.503)*	-0.00004 (3.294)**	-0.00004 (3.286)**	-0.00004 (3.183)**	-0.00004 (3.171)**	-0.00004 (3.238)**
$\sigma_{i,t-1}^R$					0.03475 (3.427)**	0.03426 (3.363)**	0.03453 (3.304)**	0.03456 (3.306)**	0.03515 (3.364)**
$ILLIQ_{i,t-1}$					0.03905 (1.909)	0.04389 (2.132)*	0.04753 (2.113)*	0.04756 (2.114)*	0.04671 (2.077)*
$BSI_{i,t-1}$					-0.00000 (1.552)	-0.00000 (1.509)	-0.00000 (1.540)	-0.00000 (1.542)	-0.00000 (1.562)
$TV_{i,t-1}$					0.00005 (3.141)**	0.00005 (3.092)**	0.00005 (2.976)**	0.00005 (2.979)**	0.00005 (2.999)**
$MT_{i,t-1}$					0.00008 (1.190)	0.00006 (1.006)	0.00006 (0.928)	0.00006 (0.925)	0.00006 (0.919)
$DEPTH_{i,t-1}$						0.00000 (0.464)	0.00000 (0.539)	0.00000 (0.538)	
$SLOPE_{i,t-1}$							-0.00004 (0.264)	-0.00004 (0.213)	
$SPREAD_{i,t-1}$								-0.00000 (0.532)	-0.00000 (0.535)
$SLOPE_{i,t-1}^B$									0.00006 (0.568)
$SLOPE_{i,t-1}^A$									-0.00009 (0.488)
$DEPTH_{i,t-1}^B$									-0.00000 (1.085)
$DEPTH_{i,t-1}^A$									0.00000 (1.884)
<i>Intercept</i>	-0.00002 (0.479)	-0.00002 (0.476)	0.00002 (0.386)	-0.00000 (0.054)	-0.00030 (3.473)**	-0.00029 (3.397)**	-0.00029 (3.325)**	-0.00029 (3.297)**	-0.00029 (3.302)**
R^2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Currency and Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	112,151	112,151	112,151	112,151	112,143	110,818	107,675	107,666	107,657
	Panel B: 60 min								
	1	2	3	4	5	6	7	8	9
$ASC_{i,t-1}$	-0.00142 (0.562)			-0.00151 (0.564)	-0.00101 (0.374)	-0.00094 (0.344)	-0.00121 (0.430)	-0.00112 (0.398)	-0.00116 (0.411)
$IC_{i,t-1}$		0.00139 (0.495)		0.00104 (0.340)	0.00153 (0.491)	0.00126 (0.403)	0.00121 (0.362)	0.00127 (0.381)	0.00124 (0.374)
$OPC_{i,t-1}$			0.00004 (0.185)	0.00005 (0.202)	-0.00025 (0.893)	-0.00025 (0.894)	-0.00028 (0.930)	-0.00026 (0.842)	-0.00026 (0.850)
$\sigma_{i,t-1}^R$					0.16354 (4.464)**	0.16330 (4.419)**	0.17015 (4.564)**	0.16997 (4.560)**	0.16968 (4.552)**
$ILLIQ_{i,t-1}$					-0.12680 (1.039)	-0.13647 (1.098)	-0.14166 (1.127)	-0.14005 (1.115)	-0.13842 (1.102)
$BSI_{i,t-1}$					-0.00000 (1.011)	-0.00000 (1.154)	-0.00000 (1.120)	-0.00000 (1.144)	-0.00000 (1.109)
$TV_{i,t-1}$					0.00011 (0.732)	0.00012 (0.775)	0.00010 (0.637)	0.00010 (0.654)	0.00010 (0.634)
$MT_{i,t-1}$					0.00175 (1.944)	0.00179 (1.973)*	0.00168 (1.836)	0.00168 (1.833)	0.00163 (1.775)
$DEPTH_{i,t-1}$						0.00002 (1.346)	0.00002 (1.222)	0.00002 (1.221)	
$SLOPE_{i,t-1}$							0.00082 (0.234)	0.00097 (0.277)	
$SPREAD_{i,t-1}$								-0.00007 (1.021)	-0.00007 (1.026)
$SLOPE_{i,t-1}^B$									0.00056 (0.162)
$SLOPE_{i,t-1}^A$									0.00046 (0.886)
$DEPTH_{i,t-1}^B$									0.00006 (1.869)
$DEPTH_{i,t-1}^A$									0.00001 (0.361)
<i>Intercept</i>	0.00036 (0.793)	0.00011 (0.229)	0.00020 (0.440)	0.00022 (0.415)	-0.00146 (1.166)	-0.00147 (1.171)	-0.00125 (0.986)	-0.00121 (0.954)	-0.00118 (0.928)
R^2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Currency and Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes